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Helium Nanodroplets: an Overview

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We review the physics of helium nanodroplets with emphasis on theoretical aspects that have been the subject of recent research, like ^{3}He and mixed ^{3}He-^{4}He droplets, vortices in droplets and droplets on weakly attractive substrates, and in others that have not been previously reviewed in detail. Whenever possible, we compare results from different theoretical methods, with the aim of assessing the capabilities and limitations of the more phenomenological ones, that often constitute the only instrument to address some of the questions raised by current experiments.

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Contents

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1. INTRODUCTION

The knowledge and understanding of helium droplets has experienced a tremendous growth in recent years that contributed, to a large extent, to keep the physics of liquid helium in particular, and of quantum fluids in general, as a lively field of research, drawing the attention of researchers from many-body theory, atomic and molecular spectroscopy, and quantum chemistry. This can be appreciated looking at the series of review articles that, starting in the early 1990's, have followed the evolution of the field both from the experimental[1–7] and theoretical[8–15] viewpoints.

There are many reasons for generating and studying helium droplets, but probably one of the main motivations has been human curiosity, as stated by Jan Northby in his review article[3]: 'let's see what happens –it's certain to be interesting and probably surprising'. As he writes, we have not been disappointed in this regard. Helium droplets share various unique characteristics with other helium systems in restricted geometries such as films on weakly attractive substrates and one-dimensional configurations. They are liquid at low pressures and zero temperature, a property not borne by any other physical system. Helium has two stable isotopes, bosonic ^{4}He and fermionic ^{3}He. Both the pure isotopes and their mixture can exist both in the thermodynamic limit as bulk liquids, and in finite selfbound samples such as droplets, with ^{3}He being the only neutral fermion with this twofold characteristic. The helium-helium interaction is weak, and the binding energies per atom are small, of the order of 7.17 K in bulk liquid ^{4}He, and of 2.5 K in bulk liquid ^{3}He. The binding energy per atom is obviously smaller in finite systems. Due to this weak cohesion, helium clusters cool down very efficiently after their formation, reaching, within microseconds, limiting temperatures (T) of around 0.38 K for ^{4}He[16], and 0.15 K for ^{3}He and mixed droplets[17]. Usually, on-flight experiments are carried out a few milliseconds after formation. Accordingly, ^{4}He drops are superfluid whereas ^{3}He ones are in the normal -non superfluid- state. It can be readily appreciated that helium clusters thus offer the possibility of studying atomic systems that obey to different quantum statistics, interact through the same atom-atom interaction, and may display superfluid effects at a nanoscopic scale.

At low temperatures, below the tricritical point of the helium mixture at about 0.87 K, the solubility of ^{3}He into bulk ^{4}He is limited[18]. This, together with the fact that the mass of a ^{3}He atom is smaller than that of a ^{4}He atom, causes that in mixed helium droplets, ^{3}He resides at the surface, coating the bosonic ^{4}He component of the nanomixture. The clusters thus reproduce the behaviour previously reported for bulk mixtures[19] and mixed films[20]. This has interesting practical and conceptual consequences. On the one hand, mixed clusters reach lower temperatures than pure ^{4}He ones. On the other hand, the bulk part of the droplet is made of pure, superfluid ^{4}He. This is useful for spectroscopic studies of complex molecules embedded into helium matrices, and has also consequences regarding fundamental studies of superfluidity at the nanoscale.

Pure helium drops are neutral systems, and this condition makes them difficult to characterize and study. Their size, composition, binding energy and excitation spectra are either unknown or known with large uncertainties, or determined in rather indirect ways. In spite of this, they have attracted the interest of theoreticians, who have investigated these properties using approaches of different complexity, depending on the size of the droplet, and on its fermionic or bosonic character. Theoretical approaches to describe $^4\text{He}_N$ doped clusters range from Quantum Monte Carlo and microscopic variational methods such as hypernetted chain/Euler-Lagrange (HNC/EL)[13], to density functional (DF) methods (see, e.g., Ref. 15). Different Monte Carlo (MC) methods, i.e., Variational, Diffusion, Green's function and Path Integral have been applied to ^{4}He systems of small and moderate size, as reviewed recently[9–14]. The more phenomenological view resorts to DF theories, which have become increasingly popular in recent years[21] as a useful computational tool to study the properties of classical and quantum fluids, especially for large systems for which they provide a good compromise between accuracy and computational cost, or for ^{3}He systems which are notoriously difficult to address with MC methods. In general, DF results compare very favorably with the more precise many-body calculations, when available, this being one of the reasons that encourages the use of these methods to board some of the problems lying beyond the scope of microscopic ones. In particular, DF has been employed to investigate very large ^{4}He drops in the presence of impurities, as well as doped ^{3}He and doped mixed clusters. Systems with rather large number of fermions can be also dealt with in the DF frame, employing the Thomas-Fermi (TF) approach.

Recently, DF theory has been applied to dynamical problems in bulk liquid ^{4}He and droplets[22–26] using the Orsay-Trento (OT) finite range density functional[27], that remarkably improves the previous Orsay-Paris (OP) functional[28]. The application of DF to bulk liquid ^{3}He, films and droplets has been recently reviewed[15], and various aspects of the emerging physical picture will be discussed later in this review.

A fundamental property of helium droplets is their ability to pick up any species with which they collide. As a consequence, they can be doped with nearly any kind of atomic or molecular species, forming new complexes. Depending on the strength of the helium-dopant interaction, the latter may reside in the bulk of the cluster, or at its surface. The possibility of doping the drops offers two complementary types of interesting experimental and theoretical studies, focusing either upon the dopant or upon the cluster.

¿From the impurity standpoint, it is possible to carry out high resolution spectroscopy studies, made possible due to the low temperature of the host. Since the helium-dopant interaction is weak, and the drops are liquid, one avoids a basic problem that arises when the intruder is isolated in the interior of crystals of rare gases or hydrogen, since in such situations, the interaction with the crystal affects drastically the geometry of the foreign molecule. Since the number of dopants in a given droplet can be controlled by changing the pressure and temperature in the pickup chamber, one may also avoid another problem often appearing in low temperature spectroscopic studies, such as molecular aggregation. Thus, it is possible to have one single molecule per doped droplet. However, one could proceed differently, taking advantage of the opportunity to control the number of captured impurities[29], and use the helium host as a matrix to create self-organized structures made of polar molecules, which otherwise would not be possible to synthetize[30,31], or very cold metal clusters whose electronic spectra can be measured by femtosecond laser spectroscopy[32,33]. One may also form metal clusters with electronic structure quite different from the respective free one, as is the case of alkali atom clusters[34,35], or study the Coulomb explosion of metal clusters grown inside He drops after they are excited by intense femtosecond laser pulses[36].

¿From the drop viewpoint, the impurity may be used as a gentle probe to access some properties that would not be determined otherwise. Two clear, closely related examples, are the superfluid character of small ^{4}He droplets[37–39], and their limiting temperatures[16,17].

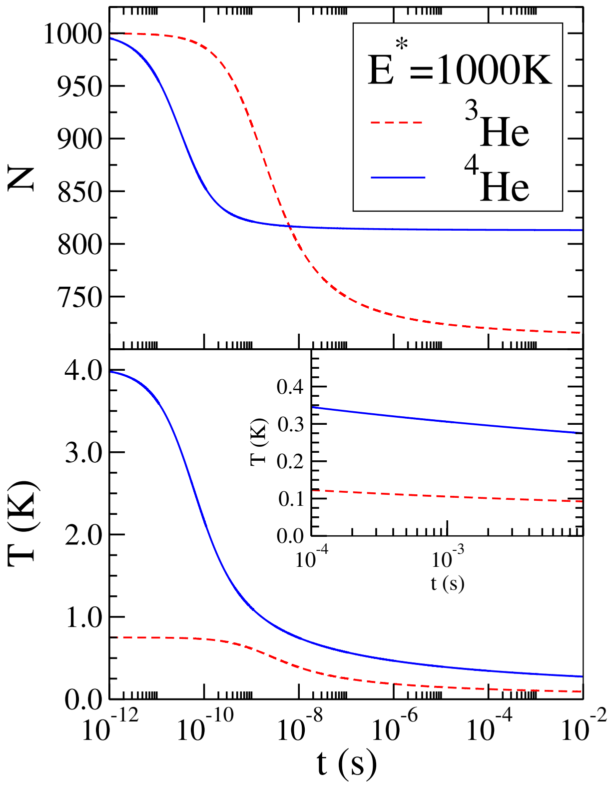

Fig. 1. (Color online) Calculated time evolution of the atom number and temperature of a ^{3}He and a ^{4}He droplet containing 1000 atoms and an initial excitation energy of 1000 K.

The experimental aspects of production, doping and analysis of helium clusters have been discussed in detail in several review articles (see, e.g., Refs. 2–7). Beams of ^{3}He, ^{4}He or mixed droplets are made by expanding helium gas at high pressure ($P = 20-100$ bar) and low temperature ($T = 5-20$

K), through a nozzle of several microns radius, into vacuum. During the expansion the gas cools down adiabatically, and depending on the stagnation chamber conditions drops of different sizes are condensed. In this process, in a few microseconds, the above mentioned limiting temperatures of 0.38 K for ^{4}He, and of 0.15 K for ^{3}He are reached. These values are in agreement with those predicted in earlier calculations[40,41], see also Refs. 42,43. Figure 1 shows the time evolution of the atom number and temperature of ^{3}He and ^{4}He droplets with initial N=1000 particles and 1000 K excitation energy[41]. This rapid cooling down of the droplets has clear advantages, as it allows to carry out on-flight spectroscopic experiments of captured atoms and molecules –see below–, but also has the disadvantage that experiments involving controlled changes in the environment temperature cannot be performed. Finite temperature calculations for ^{4}He drops in equilibrium with its vapor, some of them relevant for droplet nucleation in a saturated vapor, have been carried out in the past[44–46].

After formation, helium clusters may be deflected by a secondary beam, which allows the study of atom size distributions and scattering cross sections, if they are large enough ($N \sim 10^4$)[47,48]. For small sizes, of the order of a few tens of particles, a method based on diffraction from a transmission grating[49] has been used to identify the number of helium atoms in the system and, indirectly, the excited states of the droplets[50]. On the other hand, very large drops have been analyzed by embedding single electrons and deflecting the ionized clusters in an electric field[51]. These excess electrons serve the twofold purpose of analyzing the drops themselves on the one hand, and on the other, of studying interesting phenomena related to the dynamics of electrons inside helium clusters[52–57], especially with respect to the different behavior of superfluid ^{4}He and normal ^{3}He. The structure of electronically excited states of ^{3}He and ^{4}He clusters has been studied using fluorescence excitation spectroscopy[58–60]. To this end, the beam –consisting of helium atoms and clusters– is intersected by monochromatic tunable synchrotron radiation.

Helium clusters in the beam may be also directed across a pick-up chamber filled with vapor of dopant atoms or molecules, which are readily captured by the drops. In this process, the kinetic energy –and also any kind of internal excitation energy– of the impurity is transferred to the droplet, causing sizeable atom evaporation. The resulting complexes may be analyzed after exposure to a laser beam which incides in the opposite direction of the beam. Photon absorption leads again to large atom evaporation, decreasing further the cluster size. This may be used in depletion spectroscopy, as upon photon absorption, the electron-impact cross section for ionization is lowered and this provokes a decrease in the signal detected in a mass spectrometer.

Infrared spectra of molecules embedded into helium nanodroplets were first reported in 1992 by the Scoles group[61]. The impact of high resolution infrared molecular spectroscopy in this field has been crucial, making room to a series of studies where the superfluid character of ^{4}He has played a key role. Besides, the study of the intensity of the rovibrational spectra of embedded molecules permitted the determination of the limiting temperatures previously referred to. Spectroscopic investigations on the electronic transitions of atoms, molecules and clusters embedded in helium droplets have been also carried out, as reviewed in detail in Ref. 5. It is worth recalling that helium degrees of freedom are often excited when the impurity experiences electronic transitions, at variance with the helium response in the presence of rovibrational excitations of the dopant. This is due to the sizeable change in the helium environment surrounding the intruder –first solvation shells– upon electronic excitation. Some of these aspects will be discussed later.

In this work we review some features of the physics of helium nanodroplets with a few thousand atoms at most, leaving aside descriptions of very large systems. Experiments involving very large drops have been reviewed in Ref. 3, and some recent experimental activity on levitating neutral superfluid droplets of microscopic size is presented in Refs. 62,63. The scattering of e.g. electrons[64,65], foreign atoms[29,47], and helium atoms[13,66] off helium drops, is not explicitly considered here either, although many of the cited works necessary involve processes of this kind. Our emphasis lies on theoretical topics that have been the subject of recent research, like the ground state (gs) structure and excited states of pure and mixed ^{3}He-^{4}He clusters, vortices in droplets, and helium clusters on weakly attractive substrates.

This article is organized as follows. In Sec. 2 we shortly review some of the theoretical methods used to address the structure and excitations of helium droplets. The gs structure of the isotopically pure and mixed droplets is discussed in Sec. 3, and that of doped ones is presented in Sec. 4. The dynamical properties of helium clusters of all kinds are discussed in Sec. 5, and quantized vortices in superfluid drops are examined in Sec. 6. In Sec. 7 we analyze the physical appearance of helium clusters in the presence of extended external fields and the growth of pure and mixed droplets on planar and curved substrates; traditionally, this subject belongs to the physics of wetting, but recent experiments on surface deposition of clusters formed inside helium droplets[67], together with calculations of wetting properties from the perspective of nanoclusters condensed on a surface[68], have prompted us to address this subject in a broader perspective. Helium nanodroplets on weakly attractive substrates and their dynamics –splashing– against them have been recently addressed in Refs. 23,69. A summary is presented in Sec. 8. Finally, we point out that the words drop, cluster, nanodrop, nanocluster, etc. are used here as synonyms.

2. THEORETICAL METHODS

The theoretical methods used to study helium droplets fall into two different categories, following the use of either a bare or an effective atom-atom interaction. For simplicity, we shall refer to the former as microscopic theories. They aim at solving as accurately as possible the many-body Schrödinger equation, employing the best helium-helium interaction available. Alternatively, some selected experimental properties of the homogeneous system can be taken to construct an effective interaction, which afterwards is used in particular to describe inhomogeneous systems. As the effective interaction is density-dependent, these type of methods fall into the general framework of density functional theories. Both categories of methods are complementary, as will be shown along this paper.

2.1. Microscopic theories

The starting point is the Hamiltonian of a N-atom droplet

$$H = \sum_{i=1}^{N} \left\{ -\frac{\hbar^2}{2m} \nabla_i^2 + U(\mathbf{r}_i) \right\} + \sum_{i<j}^{N} V(\mathbf{r}_{ij}) \ , \tag{1}$$

where m is the atomic mass, $U(\mathbf{r})$ represents an external potential, and $V(\mathbf{r})$ is the interaction between helium atoms, which is independent of the mass or the spin of the atomic nucleus, and thus it is the same for both isotopes. It is characterized by a strong repulsion of the order of 10 eV at short distances (less than around 2 Å), when the electrons in the He closed-shells begin to overlap. Beyond this distance, there is a dipole-induced-dipole attraction, of around 10 K at relative distances of about 2.9 Å, and at large relative distances the potential shape has a $1/r^6-$tail, typical of van der Waals interactions. The Lennard-Jones 12-6 type potential, with parameters ϵ = 10.22 K and σ = 2.556 Å, reproduces well the attractive part, but the hard core is not realistic. More appropriate helium-helium interactions include phenomenological information such as the second virial coefficients or other properties of the gas, and some of them are also based on *ab initio* calculations. Among the most popular helium-helium interactions are the HFDHE-2[70], the HFD-B(HE)[71], and the more recent LM2M2[72] and TTY[73].

The microscopic approaches can be classified into two broad categories, Variational theories and Quantum Monte Carlo (QMC) theories. In the former, the starting point is an explicit construction of the wave function. It is useful to write it in the general Jastrow-Feenberg form[74]

$$\Psi(\mathbf{R}) = F(\mathbf{R})\ \Phi(\mathbf{R}) \ , \tag{2}$$

where the variable $\mathbf{R}$ represents the set $\{\mathbf{r}_1, \ldots, \mathbf{r}_N\}$ of $3N$ coordinates of the atoms in the drop. This ansatz is the product of a correlation function F, and a model function Φ. The latter describes the statistical properties of the system, and it includes a Slater determinant if the system contains fermions. The correlation function is usually written in terms of two- and three-body correlations

$$F(1, \ldots, N) = \exp\left\{\sum_{i<j} f_2(r_{ij}) + \sum_{i<j<k} f_3(r_{ij}, r_{ik}, r_{jk})\right\} \ . \tag{3}$$

A one-body term is more appropriately included in the model function Φ, which has the role of roughly confining the droplet. The correlation functions can be based on a 'clever guess', and written in terms of a set of parameters which are determined by minimizing the expectation value of the Hamiltonian, as done for instance in Ref. 75. Since the required multidimensional integrals are often calculated by means of Monte Carlo techniques, these approaches are named variational Monte Carlo (VMC). Alternatively, the optimal correlation functions can be directly obtained by minimizing the expectation value of the Hamiltonian, which leads to a set of coupled Euler-Lagrange (EL) equations for the functions f_2 and f_3. The Hypernetted Chain techniques permit to express the $3N$-dimensional integrals in terms of integrals containing the correlation functions[76]. The hybrid approach J-CI3 employs a fixed form for the short-range part of the two-body correlation function f_2, and linearizes the remaining part of f_2 as well as f_3, which are optimally determined[77]. More sophisticated correlations in the trial wave function are included in the so-called shadow wave function method[78,79]. A subsidiary particle (shadow) is associated to each real particle, introducing thus additional correlations between particles, not included in a Jastrow-Feenberg form. This model has been successfully employed in the study of the liquid-solid transition in liquid ^{4}He (Ref. 80).

One step beyond variational methods is the search of a direct numerical integration of the Schrödinger equation. With the imaginary time $\tau = it/\hbar$, the solution of the time-dependent Schrödinger equation can be expanded in the complete set of eigenfunctions $\{\Psi_n(\mathbf{R})\}$ of the Hamiltonian

$$\Psi(\mathbf{R},\tau) = \sum_n C_n \exp\left[-(E_n - E)\tau\right]\, \Psi_n(\mathbf{R}) \;, \tag{4}$$

where E_n is the eigenvalue associated to $\Psi_n(\mathbf{R})$. E is a trial parameter adjusted to maintain the amplitude of Ψ_0 constant, i.e. it should equal the exact gs energy E_0. The asymptotic function $\Psi(\mathbf{R}, \tau \to \infty)$ gives the exact $\Psi_0(\mathbf{R})$, provided that there is a nonzero overlap between $\Psi(\mathbf{R}, \tau = 0)$ and $\Psi_0(\mathbf{R})$. The purpose of Quantum Monte Carlo (QMC) methods is to integrate the imaginary-time Schrödinger equation, exploiting its analogy with an ordinary diffusion equation. The most popular QMC methods are the Green's Function Monte Carlo (GFMC) and the diffusion Monte Carlo (DMC) ones, which are associated with algorithms to approximate the time-independent or the time-dependent Green's function, respectively. Reviews of these methods may be found in Refs. 81 and 82 for GFMC, and in Refs. 83 and 84 for DMC.

In fact, it is difficult that a direct MC integration of the Schrödinger equation works efficiently, and it is more convenient to use the so-called importance sampling technique. It consists of introducing an auxiliary trial wave function to improve the efficiency of both GFMC and DMC methods. The common way of dealing with importance sampling in QMC is to consider the temporal evolution of a new function $f(\mathbf{R},\tau) = \Phi_T(\mathbf{R})\ \Psi(\mathbf{R},\tau)$, where Φ_T is a trial wave function previously determined. In general, the importance function Φ_T plays a triple role: to control the variance of the energy estimators with a reduced number of samples, to specify the quantum numbers and other properties of the system, and to define the set of nodal surfaces when needed.

The stochastic interpretation of the diffusion equation requires $f(\mathbf{R},\tau)$ to be a probability density, i.e. it should be semi-positive definite in the whole domain. This condition is obviously satisfied if one is considering the gs of a system of bosons, since both Ψ and Φ_T are positive definite. In the case of bosonic excited states or systems containing fermions the condition is not fulfilled. One approximation is to impose that both functions Ψ and Φ_T change sign together. This is the so-called fixed-node approximation[85–87], which has been extensively used. The nodes are imposed by the trial wave function and the nodal surface remains fixed along the calculation. It has been proved[87] that due to this constraint on the nodal surfaces, the energy is in fact a variational upper bound to the exact energy for a given symmetry.

The microscopic methods previously discussed apply to systems at zero temperature. A powerful technique to consider thermal effects is the path integral Monte Carlo method (PIMC), whose aim is to compute thermal averages of the observables of a system. It is based on the path-integral formulation of quantum mechanics by Feynman[88,89]. The temperature dependent density matrix of a quantum system is constructed, and made to evolve in imaginary time. PIMC is somehow similar to the DMC method, but with some relevant differences, namely it does not require the use of an importance sampling function and it is tailored to describe quantum systems at finite temperature. PIMC is especially well suited to address solid helium[90,91], as it does not make use of the variational ansatz of the shadow wave function method. However, it is increasingly costly from the numerical point of view as T decreases. General reviews of the method can be seen in Refs. 14,92.

The excitation spectrum of a system may be analyzed by using sum rule techniques. In fact, this method is close to the Feynman-Cohen treatment[93] of the compressional excitations in liquid ^{4}He. It has inspired a variational wave function for excited states of the form $Q(\mathbf{R})\Phi_0(\mathbf{R})$, where Q is an excitation operator and Φ_0 is either a trial or the *exact* gs wave function. The excitation energy of the first excited state is bounded by

$$E_1 - E_0 \leq \frac{M_1[Q]}{M_0[Q]} \ , \tag{5}$$

where

$$M_1[Q] = \frac{1}{2}\langle\Psi_0|[Q,[H,Q]]|\Psi_0\rangle \tag{6}$$

and

$$M_0[Q] = \langle\Psi_0|Q^2|\Psi_0\rangle - |\langle\Psi_0|Q|\Psi_0\rangle|^2 \ , \tag{7}$$

are the so-called energy-weighted and the non-energy-weighted sum rules, respectively. One is naturally led to determine the optimal excitation operator Q by minimizing the rhs in Eq. (5). This scheme has been followed by several authors to analyze the low-lying states of ^{4}He droplets. Rama Krishna and Whaley[39] used a trial gs function and parametrized a monopolar operator Q. Chin and Krotscheck[94] evaluated the gs expectation values (6-7) by a DMC calculation and optimized the operator Q to study dipolar and quadrupolar excitations.

The DMC method has been used in a different way to directly estimate the energy of the first excited state characterized by specific quantum numbers different from those of the gs[50,126]. For instance, in the case of ^{4}He drops, a simple trial wave function for states of angular momentum L is obtained by multiplying a gs ($L = 0$) trial function with the function

$$\sum_{i=1}^{N} |\mathbf{r}_i - \mathbf{R}_{\rm cm}|^L P_L(\cos\bar{\theta}_i) \ , \tag{8}$$

where $\bar{\theta}_i$ is the azimuthal angle of the vector $\mathbf{r}_i - \mathbf{R}_{\rm cm}$ which defines the position of the ith particle with respect to the center-of-mass coordinate $\mathbf{R}_{\rm cm} = \sum_i \mathbf{r}_i/N$, and P_L is the Legendre polynomial. Although from a variational viewpoint this is not the best trial wave function, it provides the relevant importance sampling in a DMC calculation. Indeed, the initial state at $\tau = 0$ is an admixture of eigenstates of a specified angular momentum L. The lowest eigenstate in the subspace of angular momentum L is selected along the time evolution, the remaining contributions being exponentially suppressed as the time increases. Since the trial wave function is not positive definite, the fixed-node approximation has been applied, and consequently an upper bound to the energy of the first excited state with angular momentum L is obtained.

2.2. Density functional theories

Phenomenological methods to study properties of helium liquids started with the celebrated theories by Landau of the low-lying excitations of liquid ^{4}He[95], and the Landau-Fermi theory of ^{3}He[96]. In what concerns finite systems, the simplest phenomenological approach is the liquid droplet model (LDM), originally developed to obtain in a simple way the gs energies of atomic nuclei and their collective excitations[97]. In the LDM spirit, the gs energy is written as an expansion in powers of the number of atoms in the drop N, with the leading coefficients identified as bulk, surface, and curvature energy. To describe the collective modes, sharp surface droplets of constant density are considered, whose total energy is given in terms of collective variables representing the surface or volume deviations from the equilibrium gs. In the limit of large drops, the solution of the hydrodynamical equations with appropriate boundaries allows one to get the energies of collective states related to bulk and surface excitations, see e.g. Ref. 8.

Models to investigate surface properties of superfluid ^{4}He were developed during the seventies. At zero temperature and in the absence of currents, the superfluid order parameter is simply the square root of the particle number density and it becomes natural to think of a density functional (DF) to describe superfluid helium[98,99]. In fact, the theoretical support of DF theory is the Hohenberg-Kohn theorem[100], which guarantees that the gs energy of a many-body system is a unique functional of its density. Moreover, the Kohn-Sham (KS) method[101] allows one to obtain from this functional not only the gs properties but also the elementary excitation spectrum. However, this theorem does not provide any clue for the construction of the functional, which has to be written down resorting to physical insight and phenomenological approaches.

The DF can be written in the general form

$$E[\rho] = \int \mathrm{d}\mathbf{r} \left\{ \frac{\hbar^2}{2m} \tau[\rho(\mathbf{r})] + \varepsilon_c[\rho(\mathbf{r})] + \rho(\mathbf{r})\, U(\mathbf{r}) \right\} , \tag{9}$$

where $U(\mathbf{r})$ is an external potential, $\varepsilon_c[\rho]$ is the correlation energy density, and $\tau[\rho]$ is the kinetic energy density of the corresponding non-interacting system. In the case of ^{3}He systems, the atomic mass m is often replaced by a density-dependent effective mass m^*. Once written the functional, the total energy is minimized with respect to the Kohn-Sham single-particle (sp) orbitals ϕ_α, which define the particle and kinetic energy densities. This leads to mean field equations which have to be solved self-consistently to determine the sp orbitals ϕ_α and energies ϵ_α. In the case of ^{4}He atoms, if the system is fully condensed, there is just a single orbital, and it is more convenient to minimize the energy with respect to the density, replacing the kinetic energy density with the exact expression $\tau = |\nabla \sqrt{\rho(\mathbf{r})}|^2$.

The Stringari-Treiner (ST) functional[102] is the first and simplest DF. However, due to its zero-range character, it lacks the asymptotic $1/r^6$ tail of the He-He interaction. Even more important, zero-range functionals miss the fact that the fluid has a characteristic microscopic length, the radius of the interaction repulsive core. This renders them useless to describe situations where the liquid is probed on a microscopic scale, e.g., near the core of a quantized vortex, or close to an impurity. Improved functionals that include these effects have been first proposed in Ref. 28 for ^{4}He (OP functional), and in Refs. 103–105 for ^{3}He. They include a Lennard-Jones potential term, conveniently screened at short distances, and also a coarse-grained density, as in the study of classical liquids[106]. A DF for ^{4}He systems was latter constructed (OT functional[27]) to describe the static and dynamical response of the homogeneous liquid. This functional improves by construction the description of properties such as the momentum dependence of the static response function and the phonon-roton dispersion relation. Besides, it is remarkable that both OP and OT functionals predict a surface tension of the liquid at zero T in very good agreement with experiment. Very recently, a fully variational DF method has been proposed[107] that can accurately describe the solid phase of ^{4}He and the freezing transition of liquid ^{4}He at zero temperature. This opens the possibility to use unbiased DF methods to study highly non-homogeneous systems, like ^{4}He drops doped with very strongly attractive impurities or substrates. The process of fixing a density functional for liquid ^{3}He is much more involved because spin properties have to be also considered. Some attempts have been made in Refs. 108–110

In the case of systems formed by a mixture of both isotopes, one considers the total energy as a DF of the particle densities ρ_4 and ρ_3 of each isotope. A zero-range functional for homogeneous mixtures was thus constructed[111] to study the properties of the solution as a function of the relative concentration of the isotopes as well as the interface between both liquids for homogeneous mixtures. Inspired in the OT functional for ^{4}He, a finite range one was proposed in Ref. 112, which reproduces various experimental quantities for varying relative concentrations and pressures.

The structure of most DF's allows one to obtain with some effort, the spectrum of pure helium clusters within the random-phase approximation (RPA). This approach was used for the first time in the context of ^{4}He drops by Casas and Stringari[113]. Using the ST functional, they obtained the energies of the volume monopole mode, and of the surface quadrupole and octupole modes. Furthermore, they carried out a sum-rules analysis of these excitations. To our knowledge, this is the first microscopic calculation of the collective modes of $^4\mathrm{He}_N$ drops, that previously had been only worked out using classical, LDM expressions as, for example, that corresponding to the ripplon modes of multipolarity $L \geq 2$:

$$\hbar\omega_L = \left[\frac{4\pi}{3}\frac{\sigma}{m} L\,(L-1)\,(L+2)\right]^{1/2} N^{-1/2} \;, \tag{10}$$

where σ is the surface tension of the liquid and m is the mass of a helium atom. For ^{4}He, the LDM expressions work rather well compared to RPA results, except for drops containing less than $\sim$ 100 atoms in the case of ripplon modes, and less than $\sim$ 500 atoms in the case of the bulk monopole mode[114].

In RPA frame, the elementary excitations of the system are given by the poles of the density-density Green's function, which can be found by solving the integral equation

$$\begin{aligned} G^{RPA}(\mathbf{r}_1,\mathbf{r}_2,\omega) &= G^0(\mathbf{r}_1,\mathbf{r}_2,\omega) \\ &+ \int d\mathbf{r}_3\, d\mathbf{r}_4\; G^0(\mathbf{r}_1,\mathbf{r}_3,\omega)\; V^{ph}(\mathbf{r}_3,\mathbf{r}_4)\; G^{RPA}(\mathbf{r}_4,\mathbf{r}_2,\omega). \end{aligned} \tag{11}$$

In this equation, $G^0(\mathbf{r}_1,\mathbf{r}_2,\omega)$ is the Hartree (^{4}He case) or Hartree-Fock (HF) (^{3}He case) Green's function, and V^{ph} is the particle-hole interaction. In the case of ^{4}He, G^0 has been obtained either from its spectral decomposition in terms of sp energies and wave functions[113,115], which implies a discretization and truncation of the Hartree spectrum, or by a more elaborate method, called continuum RPA that avoids introducing a truncation[114]. This latter method is especially well suited for high energy excitation modes, like the response of the droplets to the plane wave operator in the high-q limit, or when a sizeable part of the strength, for a given L, is in the continuum, i.e., above the atom emission threshold energy.

Formally, the particle-hole interaction V^{ph} is given by the second functional derivative of the total energy with respect to densities taken at the Hartree or HF solution. Expressions for V^{ph} whose complexity depends on the DF employed, can be found in Refs. 113–115 for ^{4}He, and in Refs. 116,117 for ^{3}He. The solution of the RPA integral equation is greatly simplified for spherically symmetric droplets. In this case, V^{ph} and G^{RPA} admit a straightforward multipolar decomposition and the resulting equations can be solved separately for each multipole.

Within DF theory, a very powerful method has been recently implemented to calculate the response of helium droplets to external probes that allows one to handle interesting physical situations such as helium systems in confined geometries, and in general, deformed helium systems, in an easier way than either previous ones. It is based on the solution of the time-dependent DF (TDDF) equations obtained from the application of a variational principle to a suitable action. The method, often called real-time RPA, has been also applied to obtain the spectrum of simple metal clusters and of quantum dots[118,119].

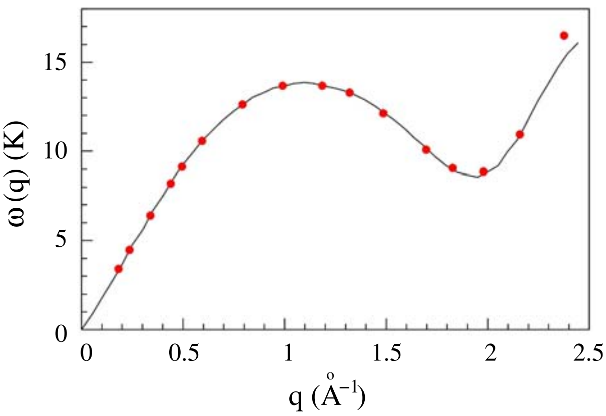

Fig. 2. (Color online) Bulk ^{4}He excitation spectrum $\omega(q)$. The points are the calculated spectrum[23], and the solid line shows the experimental dispersion relation (by courtesy of F. Ancilotto).

In the ^{4}He case, one starts from an action integral $\mathcal{A}$ written in terms of a complex 'wave function' $\Psi(\mathbf{r}, t)$

$$\mathcal{A}[\Psi] = \int dt d\mathbf{r} \left\{ \mathcal{E}[\Psi, \Psi^*] - i\hbar \Psi^* \frac{\partial \Psi}{\partial t} \right\} , \tag{12}$$

where $\mathcal{E}$ is the energy density, $\int d\mathbf{r}\, \mathcal{E} = E[\rho, \mathbf{v}]$. Writting $\Psi(\mathbf{r}, t) \equiv \phi e^{i\Theta}$ one gets the time-dependent density $\rho(\mathbf{r}, t) = \phi^2$ and the fluid velocity field

$\mathbf{v}(\mathbf{r},t) = \frac{\hbar}{m}\nabla\Theta$. The velocity dependence stems from terms introduced to phenomenologically account for backflow effects[27].

To obtain the excitation spectrum of a ^{4}He system, one first solves the static problem to determine the gs $\Psi(\mathbf{r})$. This wave function is then modified usually multiplying it by a perturbation field whose functional form may allow to excite the selected mode, and it is let to evolve in time using the TDDF equation. The ability of the method has been shown in Ref. 23. Fig. 2 shows that the numerical implementation of the TDDF method is accurate enough to reproduce the elementary excitations of bulk liquid ^{4}He, which was an input to build the Orsay-Trento DF used in this calculation. In general, to solve accurately the TDDF equation is a very demanding numerical task. Examples of other applications will be presented in subsequent Sections.

3. THE STRUCTURE OF HELIUM DROPLETS

In this Section we shall summarize the main results about the gs of pure ^{4}He and ^{3}He droplets, as well as of mixed droplets of both isotopes.

3.1. Droplets of ^{4}He atoms

They have been theoretically studied with a variety of methods, based either on microscopic interactions[39,75,77,94,120–126] or density functionals[102,114]. QMC methods allow the exact evaluation of the gs energy of bosonic systems at zero temperature. However, one should keep in mind that the helium-helium interaction is not unambiguously determined, and the predicted energies depend on the specific interaction employed. To illustrate this point we display in Table 1 the results obtained for liquid ^{4}He with the DMC calculations of Ref. 84, and for the dimer ^{4}He$_2$ with a direct integration of the Schrödinger equation.

Table 1. Exact results obtained with different He-He interactions.

	Liquid ^{4}He		Dimer ^{4}He$_2$	
Interaction	$-E_0/N$ (K)	ρ_0 (Å^{-3})	$-E_0$ (mK)	$\langle r\rangle^{1/2}$ (Å)
HFDHE-2	7.103	0.02159	0.833	84.6
HFD-B(HE)	7.277	0.02184	1.691	61.7

The experimental results of Ref. 127 for liquid ^{4}He are $E_0/N = -7.17$ K and $\rho_0 = 0.02185$ Å^{-3} -saturation density. The HFD-B(HE) interaction reproduces the equilibrium density, but slightly overbinds the liquid. However, these authors indicate[127] that the use of a more recent temperature scale would yield $E_0/N = -7.23$ K, closer to the HFD-B(HE) results. It is also worth noticing that DMC calculations with this interaction reproduce remarkably well, as a function of pressure, the experimental values of the compressibility and the speed of sound at equilibrium[84].

Any number of ^{4}He atoms at zero temperature forms a self-bound system. Experimental indication of the dimer $^4He_2$ was reported in Ref. 128 using electron impact ionization techniques. The diffraction technique of Ref. 129 has provided conclusive evidence of its existence[130] and also leads to an estimation[131] of its bond length, 52 ± 4 Å, and indirectly of its binding energy, 1.1+0.3-0.2 mK. In Table 1 are given the energies and root mean square (rms) radii of $^4He_2$ obtained with two interactions. It can be seen that the difference of nearly 0.17 K between the energies per particle in the bulk obtained with HFDHE-2 and HFD-B(HE) interactions gives rise to a difference of around 0.8 mK in the dimer, roughly the total energy calculated with HFDHE-2. The relatively small differences provided by the different interactions can be magnified in calculations of energies of ^{3}He or mixed ^{4}He-^{3}He systems at the threshold of stability discussed in Sec. 3.3.2.

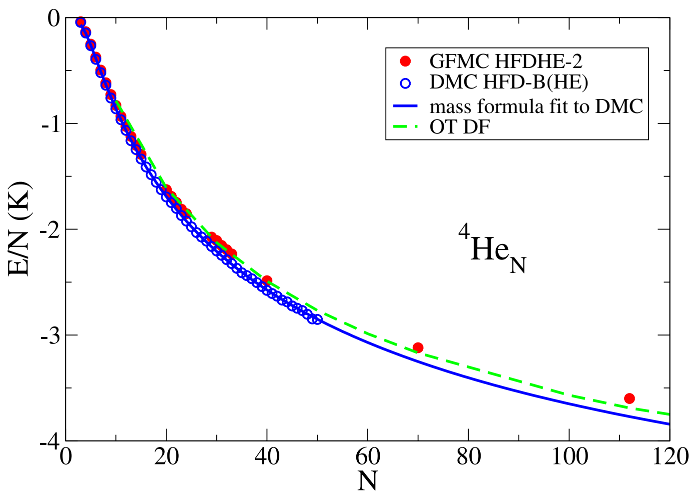

Fig. 3. (Color online) Ground state energies per particle of a ^4He_N drop as a function of the number N of atoms, calculated with several interactions and methods.

The energy per particle for a series of helium droplets is plotted in Fig 3. Filled circles correspond to the GFMC calculations with the HFDHE-2 interactions[120,121]. The open circles are the DMC results of Ref. 126 with the HFD-B(HE) interaction. The difference of about 0.17 K for E_0/N in the liquid given by the two interactions is reflected in the droplets. The dashed line in the figure corresponds to the results of a DF calculation with OT, which employed the value $E_0/N = -7.15$ K in the fitting of its parameters. The solid line is a mass formula fit to the DMC results, obtained for $N \leq 50$, fixing the volume coefficient to the DMC result for the liquid, and the surface one to the experimental value[132]. The full fit is

$$E/N = -7.277 + 16.96N^{-1/3} + 6.978N^{-2/3} - 41.75N^{-1} + 26.15N^{-4/3} \quad (13)$$

with all coefficients given in K. The value $a_s = 16.96$ K of the surface coefficient is obtained from the experimental surface tension[132] $\sigma_4 = 0.274$ KÅ^{-2} as $a_s = 4\pi r_0^2 \sigma_4$, where r_0 is the bulk radius such that $4\pi r_0^3 \rho_0/3 = 1$.

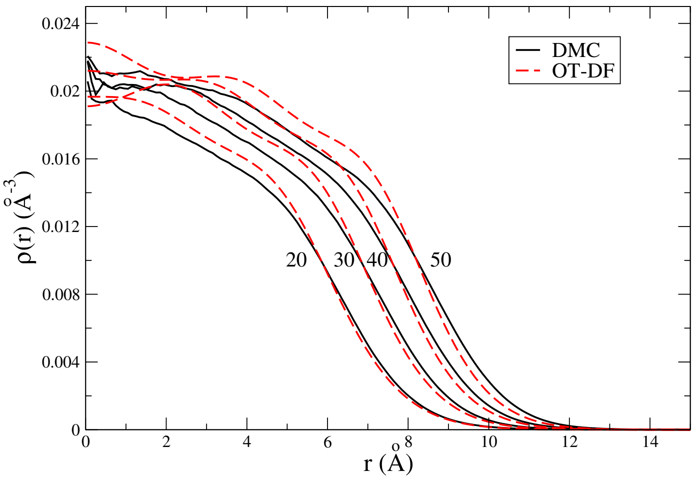

Fig. 4. (Color online) Density profiles of some ^{4}He drops. Comparison between DMC and OT-DF results.

In Fig. 4 are compared the density profiles of a few droplets, calculated with the DMC method and the OT density functional. Each density is normalized to the number of atoms. The DMC results (solid lines) have large statistical errors near the origin, which in the smoothed curves of Fig. 4

appear as a quite erratical behavior at small distances. An interesting fact is the presence of radial oscillations. They were first observed in GFMC calculations[120] for $N = 70$, and in DMC calculations[94] for $N = 70$ and 112. No oscillations appear when the DF is zero-range[102]. Although it was suggested long ago[133] that ripplons could manifest into such oscillations, its origin remains unknown at present. The authors of Ref. 120 considered these oscillations to be a spurious result related to the correlations along the stochastic random walk, but this interpretation has been ruled out by subsequent microscopical and DF calculations. One could think that such oscillations reflect an underlying quasi-solid structure which, although not included in the importance sampling driving function, the DMC evolution could be able to magnify. The OT functional predicts similar oscillations, which have been related[27] to the fact that it also reproduces the experimental peak of the bulk static response function at the roton wavelength.

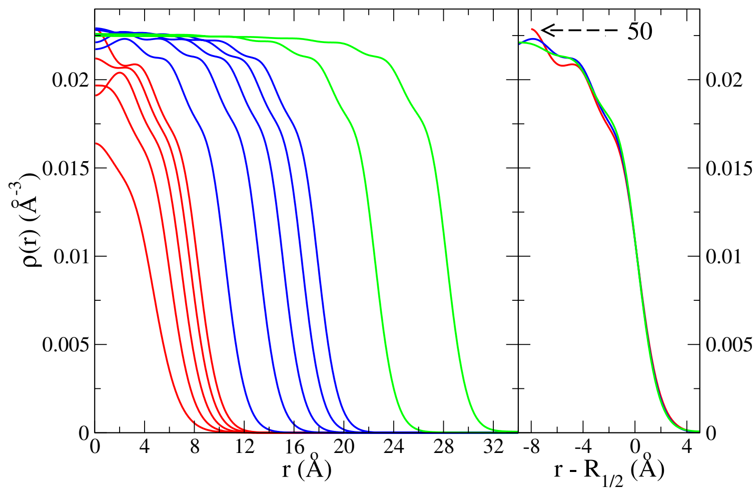

Fig. 5. (Color online) Density profiles as calculated with the OT-DF for several ^{4}He droplets. On the left panel, the number of atoms go from 10 to 50 in steps of 10 (red curves), from 100 to 500 in steps of 100 (blue curves), and 1000 and 2000 (green curves). In the right panel, the density profiles for droplets with 50, 100, 500 and 2000 atoms are represented after shifting the origin to the point $R_{1/2}$ defined as $\rho(R_{1/2}) = \rho_0/2$, where ρ_0 is the bulk density.

In Fig. 5 are displayed the density profiles obtained with the OT functional, for ^{4}He droplets with N between 10 and 2000. The radial oscillations are persistent and near the surface they are the same for all droplets with $N \geq 50$. This is illustrated in the right panel of Fig. 5, where we represent some of these profiles shifting the origin of coordinates to the point $R_{1/2}$ at which the value of the density for each droplet is equal to half the saturation density. Consequently, the density profiles for $N = 50$ does not continue up to the left end. It can be seen in this panel that all droplets have the same density profile in the surface region. The distance between the two density shoulders at the surface is about 2.8 Å, sensibly related to the value of the hard core radius, giving support to a kind of soft sphere packing as the origin of these oscillations.

The surface thickness, defined as the distance between the radial points where the density has decreased from the saturation value by 0.1 and 0.9, is clearly the same for all droplets with $N \geq 50$, and its value is about 5.2 Å. Measurements for ^{4}He films on polished Si wafers cut along the (111) plane at 0.45 K[134] yield values that vary from 5.3 ± 0.5 Å for 36 ± 1.5 Å thick films to 6.5 ± 0.5 Å for 125 ± 1.5 Å thick films.

3.2. Droplets of ^{3}He atoms

Contrary to the ^{4}He case, a minimum number of ^{3}He atoms is required to form a self-bound system, as a consequence of both the larger zero-point motion and the Pauli effect. To elucidate the relative importance of these effects, four systems with eight helium atoms were studied in Ref. 75, assuming either ^{4}He or ^{3}He mass, and either Bose or Fermi statistics. These calculations show that both the ^{3}He bosonic and the ^{4}He fermionic systems are weakly bound, whereas the ^{3}He fermionic system is unbound. Both zero-point motion and statistics have a similar effect on the binding.

Theoretical calculations of ^{3}He droplets indicate that twenty ^{3}He atoms are not enough to form a bound system, but 40 atoms are bound[75,102,104,105]. The threshold number N_{min} of ^{3}He atoms to form stable droplets is therefore in the interval between 20 and 40 atoms. The choice of these particular numbers of atoms is related to the fermionic shell-closure. In Fig. 6 are plotted the sp levels of ^{3}He$_N$ droplets, obtained from the self-consistent solution of the KS equations with the DF of Ref. 135. Up to 168 atoms, the magic numbers correspond to the harmonic oscillator scheme: they are given by the sequence $(p+1)(p+2)(p+3)/3$, with p a non-negative integer number, which gives the values 2, 8, 20, 40, 70, 112, 168, 240, 330, ... Beyond 168 atoms, the sequence of magic numbers depends sensibly on the specific density functional[102,104,105,110]. It is worth noticing that from the microscopic viewpoint, nothing is known about the ordering of the shells, which are an input in the calculations, usually assumed to be given by the harmonic oscillator sequence. In contrast, DF calculations do predict the above mentioned harmonic oscillator filling scheme. We shall come back to this point later on.

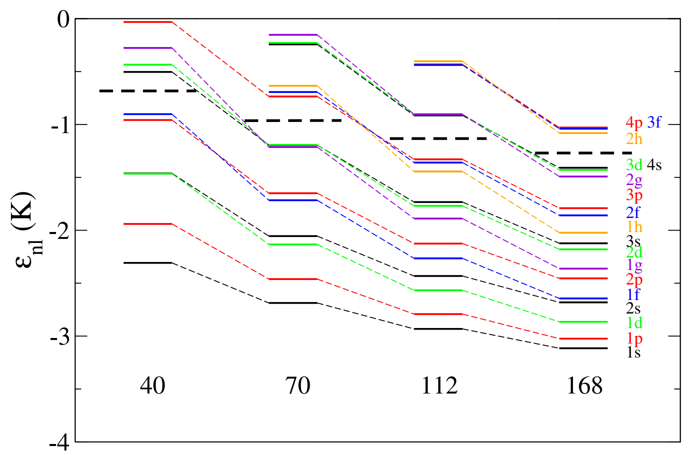

Fig. 6. (Color online) Single-particle levels in $^3\mathrm{He}_N$ droplets with $N = 40$, 70, 112, and 168, obtained by solving the KS equations with the functional of of Ref. 135. In each case, the horizontal dashed lines separate occupied from unoccupied levels The value of the sp orbital angular momentum $l = 0, 1, 2, 3, \ldots$ i.e., $s, p, d, f, \ldots$ is also indicated.

N_{min} was calculated in Ref. 136, by using the BHN non-local finite-range density functional[110]. The KS procedure provided the sp orbitals as well as the residual interaction required to perform a configuration interaction calculation in the open $(1f2p)$ harmonic oscillator shell. This DF-plus-shell-model procedure gave the stability threshold at 29 atoms, although it was argued that as far as the DF could be affected by some uncertainties, it should be safer to fix the value $N_{min} = 30$. Sometime after the publication of Ref. 136, the evaluation of the correlation energies displayed in Fig. 1 of that paper was improved. As a result, they were found to be $\sim 30\%$ smaller than the published ones. Using the improved correlation energies, it turns out that, indeed, $N_{min} = 30$. All the other conclusions of that work remain unchanged.

A J-CI3 variational microscopic study based on the HFD-B(HE) interaction was carried out in Refs. 137,138. The product of two Slater determinants, one for each spin orientation, takes care of the required fermionic antisymmetry of the wave function. The harmonic oscillator filling was assumed, and it was concluded that the threshold number is less than or equal to 34 atoms. Indeed, due to the variational character of the computational procedure, only an upper bound was obtained. This estimate was improved in Ref. 139 by means of the fixed-node diffusion Monte Carlo (DMC) procedure, for selected systems near this value. Several shell-model orderings were considered, probing thus in practice different nodal surfaces according to the various configurations employed. The results indicate that the $2s$ subshell is more deeply bound than the $1f$ subshell, i.e. the harmonic oscillator ordering seems to be preferred. The upper bound $N_{min} \leq 32$ was established in this way. The precise determination of N_{min} should require a calculation beyond the variational fixed-node approximation. It is worth stressing that as far as the binding energy of the threshold droplet is very close to zero, the predicted value for N_{min} could be very sensitive to the He-He interaction employed in the calculation, as mentioned in the previous subsection.

An interesting outcome of these calculations is the structure of the droplets with $N = 20$ to 40. The DF-plus-shell-model calculations showed that their gs have the maximum spin value allowed by Pauli's principle, i.e., $\mathrm{S}_{\mathrm{max}} = \tilde{n}/2$, where $\tilde{n}$ is the number of particles (holes) in the valence space below (above) midshell. Similar results were obtained in VMC calculations[137,138]. In Ref. 137 the determinants for each spin orientation were constructed in a Cartesian basis, so that the trial wave functions have not well-defined orbital angular momentum. In Ref. 138, the trial wave functions do have good orbital angular momentum, but the occupied states are restricted to a single valence subshell (either $1f$ or $2p$) in the $p = 3$ harmonic oscillator shell. In both cases no configuration mixing between different subshells was considered. In both calculations, it turned out that the gs is characterized by having the maximum allowed spin, and that for a given spin, states with different values of the total angular momentum L were nearly degenerate.

This result can be understood by looking at the properties of the residual interaction. The DF calculations of Ref. 136 indicate that the matrix elements in a single shell are very repulsive for $L = 0$, essentially zero for even $L-$values, and attractive, and roughly constant, for odd $L-$values. The diagonal matrix elements involving two l-subshells are much more attractive in the $S = 1$ channel than in the $S = 0$ one. Moreover, the $S = 1$ matrix elements are nearly $L-$independent. As a consequence, the energy minimum within one l-shell is obtained for configurations maximally antisymmetric in orbital space which means maximally symmetric in spin space. In addition, the aligned states of two l-subshells couple to maximum total spin due to the dominance of the spin vector channel in the cross shell interaction. An almost exact degeneracy with the orbital angular momentum was also found. The VMC calculations of Ref. 137 confirm qualitatively this behavior of the matrix elements.

The gs energies per particle can be properly collected as a mass formula. For instance, the DF results of Ref. 104 give

$$E/N = -2.49 + 8.51N^{-1/3} + 3.12N^{-2/3} - 20.4N^{-1} \tag{14}$$

with all coefficients given in K. The value $a_s = 8.51$ K of the surface coefficient is in agreement with the experimental surface tension[140] $\sigma_3 = 0.113$ KÅ^{-2}, as $4\pi r_0^2 \sigma_3 \simeq a_s$.

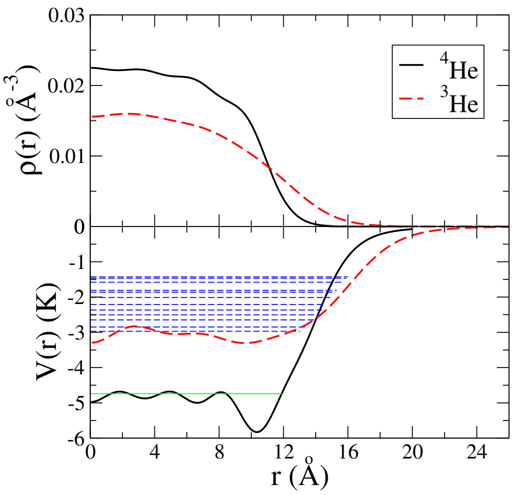

Fig. 7. (Color online) Density profiles (upper panel) and mean fields (lower panel) for $^4He_{112}$ and $^3He_{112}$ droplets. In the lower panel, the horizontal dashed lines represent the energies of the occupied ^{3}He sp levels, and the horizontal solid line represents the ^{4}He chemical potential.

In Fig. 7 are given the density profiles and mean fields for the droplets $^4He_{112}$ and $^3He_{112}$, as a typical case. The calculations for ^{4}He have been carried out with OT, and with the functional of Ref. 135 for ^{3}He. The weaker potential well of ^{3}He droplets originates a larger surface width in the density profile, typically 1 Å larger than that of a ^{4}He droplet with the same number of atoms[47,48]. The ^{3}He density profile also displays oscillations due both to shell effects inherent to its fermionic character, and to its quasi-crystal structure, as in ^{4}He drops. Other functionals[104] give rise to more pronounced density oscillations, with a spatial distance between peaks of about 4.5 Å. We recall that correlation effects, which are absent in the DF mean field description, smooth out these oscillations. In particular, the VMC ^{3}He density profiles[75] show less structure.

Finally, we present in Fig. 8 the density profiles of several ^{3}He droplets obtained with the DF of Ref. 135. In the case of $N = 40$, the dots represent the fixed node DMC calculations of Ref. 141. The energy per atom of ^{3}He$_{40}$ is slightly larger within DF, -0.162 K, as compared with the DMC value $-0.159(2)$ The mass formula Eq. (14) yields too much energy, -0.245 K.

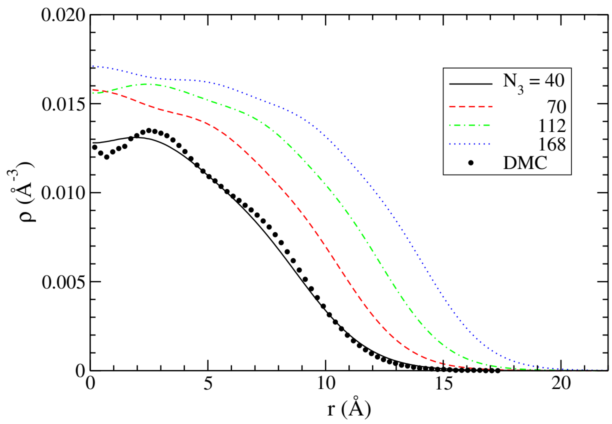

Fig. 8. (Color online) Density profiles for ^{3}He$_N$ droplets with $N = 40$, 70, 112, and 168. The dots along the $N = 40$ profile represent the fixed node DMC result of Ref. 141

3.3. Mixed ^{4}He+^{3}He droplets

The study of isotopically mixed systems at low T is very appealing. They are made of bosons and fermions with different mass interacting through the same potential, and quantum effects due to statistics and to the zero-point motion of each isotope are crucial. The surface tension of liquid ^{4}He is substantially lowered when a small amount of ^{3}He atoms is added because the larger zero-point amplitude of the latter pushes them to the free surface, where they occupy the so-called Andreev states[142], characterized by quasi two-dimensional wave functions. We shall consider now mixed droplets $^4\mathrm{He}_{N_4}$+$^3\mathrm{He}_{N_3}$ with variable number N_4 and N_3 of bosons and fermions. They will be referred to as (N_4, N_3).

3.3.1. *Mixed droplets with a single ^{3}He atom*

The simplest 'mixed' system is formed by a single ^{3}He atom attached to a $^4\mathrm{He}_N$ cluster, and its properties have been investigated using several theoretical methods. The first systematic study of the structure and energetics of $^4\mathrm{He}_N{}^3\mathrm{He}$ clusters was made by Dalfovo[143], based on a zero-range density functional. These calculations were later repeated using a non-local finite-range functional[112], yielding small quantitative differences.

The Lekner approximation was used in the VMC calculation of Ref. 144 as well as in the HNC-EL calculation of Ref. 13. Such an approximation[145], introduced to analyze the Andreev states of ^{3}He in bulk ^{4}He, assumes that the correlations of pairs ^{3}He$-^4$He and ^{4}He$-^4$He are the same. The Lekner approximation was relaxed in the DMC calculations of Refs. 146–149 on small $^4\mathrm{He}_N{}^3\mathrm{He}$ clusters.

In order to classify the resulting spectra of the ^{3}He atom, Krotscheck and Zillich[13] defined an effective wave number $k = \sqrt{\ell(\ell+1)}/R$ for each excitation characterized by the orbital angular momentum ℓ, where R is the equivalent hard sphere radius $R = \sqrt{5/3}r_{\rm rms}$, defined in terms of the rms radius $r_{\rm rms}$ of the droplet. By plotting all excitation energies as a function of k for a large number of clusters, Krotscheck and Zillich found that all results fall reasonably well on a universal quadratic line.

This result can be easily explained in a DF description. The ^{4}He density ρ_4 and the ^{3}He orbitals $\phi_{n\ell}$ are the solution of a system of coupled equations. Assuming spherical symmetry, it is written as

$$\left[-\frac{\hbar^2}{2m_4}\left(\frac{d^2}{dr^2}+\frac{2}{r}\frac{d}{dr}\right)+V_4(r)\right]\sqrt{\rho_4(r)} = \mu_4\sqrt{\rho_4(r)} \tag{15}$$

$$\left[-\frac{\hbar^2}{2m_3^*}\left(\frac{d^2}{dr^2}+\frac{2}{r}\frac{d}{dr}\right)-\frac{d}{dr}\left(\frac{\hbar^2}{2m_3^*}\right)\frac{d}{dr}\right]\phi_{n\ell}(r) \tag{16}$$
$$+\left[V_3(r)+\frac{\hbar^2}{2m_3^*}\frac{\ell(\ell+1)}{r^2}\right]\phi_{n\ell}(r) = \varepsilon_{n\ell}\phi_{n\ell}(r)$$

In these equations, μ_4 is the ^{4}He chemical potential and $\varepsilon_{n\ell}$ is the sp energy of the ^{3}He atom in the state characterized by the radial and orbital angular-momentum quantum numbers n and ℓ, respectively. The effective potentials V_4 and V_3, as well as the effective mass m_3^* do depend on ρ_4.

As the size on the drop increases, the ^{3}He atom is pushed to the surface region, due to its large zero-point motion, and for large enough clusters, the centrifugal term $\ell(\ell+1)/r^2$ entering the equation for the ^{3}He atom can be treated as a perturbation. The emerging picture is that of a molecular rigid rotor formed by the ^{4}He$_N$ cluster and the ^{3}He atom, tied by a spring with a rather large rigidity constant[143].

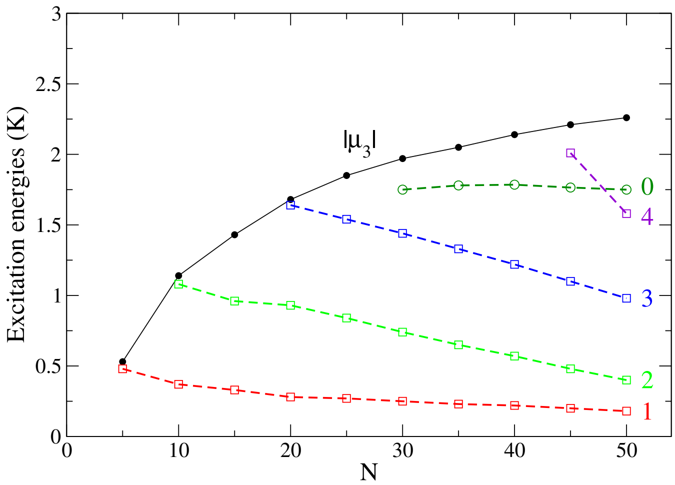

Fig. 9. (Color online) Excitation energies of the ^{3}He atom in the ^{4}He$_N$ cluster as a function of the number of atoms N. The angular momentum ℓ is indicated in the right part of each curve. The absolute value of the ^{3}He chemical potential μ_3 is also displayed as a function of N.

The DMC excitation energies of the ^{3}He atom calculated in Ref. 150 are displayed in Fig. 9, as well as the ^{3}He chemical potential, defined as the difference between the total energies of the drops ^{3}He^{4}He$_N$ and ^{4}He$_N$. We anticipate that for these 'mixed' droplets, μ_3 is what we should call in Sec. 4 'solvation energy' of a ^{3}He atom into a ^{4}He droplet, see e.g. Fig. 24.

The excitation energy of a long-lived excited state should lie below $|\mu_3|$. It can be seen that as N increases more excited levels become long-lived. The general trend of the spectrum is a series of tight rotational bands on top of radial excitations, related to increasing the number of nodes in the radial wave function. In the limit of very large N, the spectrum of the ^{3}He atom becomes independent of the quantum angular momentum, forming the analogous of the two-dimensional Andreev states in bulk helium. One may also compute 'monopolar excitations' in which, for a given ℓ-level, the radial quantum number n is increased. It can be seen that in this range of N values, only the 'monopolar excitation' corresponding to the $\ell = 0$ level is bound. Similar qualitative results have been obtained within DF theory[112,143].

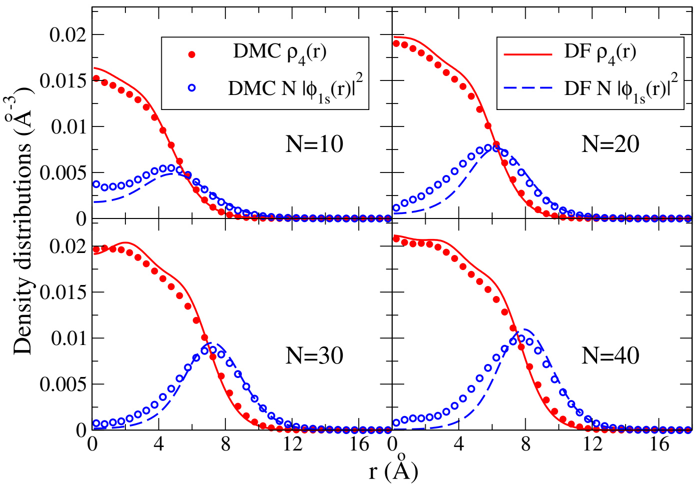

Fig. 10. (Color online) Boson density profiles (filled circles and solid lines) and fermion probability densities (open circles and dashed lines) in mixed $^4He_N{}^3He$ droplets. Each fermion probability density has been multiplied by N. Circles and lines correspond to DMC and DF calculations, respectively.

The density distributions ρ_4 and $|\phi_{1s}|^2$ are displayed in Fig. 10 for the gs of drops with $N = 10$, 20, 30 and 40. In the figure are plotted the DMC and DF results of Ref. 150 and 112, respectively. As expected, the ^{3}He atom is always located at the surface of the drops, with an appreciable probability of being at its center for the smaller N. Increasing the value of the angular momentum l produces the fermion to leave the boson cluster. It can be seen that both DMC and DF results are in qualitative agreement, with the ^{3}He atom more concentrated at the surface in the latter case.

3.3.2. Small mixed droplets

The study of the smaller mixed systems is a challenge for theoretical methods because, due to their very small binding energy, some of them are at the stability threshold. There are several interesting questions concerning the smaller mixed droplets. Which is the minimum number of ^{4}He atoms able to stabilize an otherwise unbound ^{3}He system below $N_{\rm min}$? How many ^{3}He atoms can be bound by a droplet with few ^{4}He atoms? Do all (N_4, N_3) combinations correspond to stable systems?

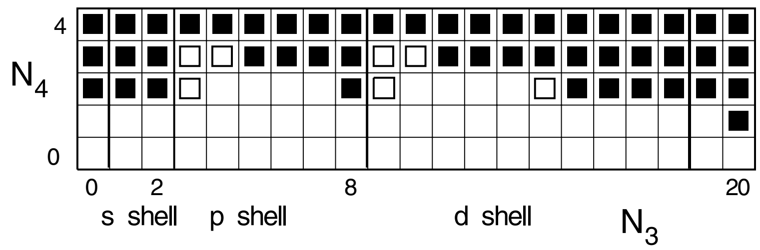

Fig. 11. Stability chart of mixed $^4{\rm He}_{N_4}$+$^3{\rm He}_{N_3}$ droplets. Filled (open) squares indicate stable (metastable) combinations. No symbol corresponds to unbound systems. The fermionic shells are indicated with the usual spectroscopic notation.

In Ref. 151 were reported variational calculations of systems containing up to five atoms of the two isotopes. A systematic study of the stability of systems with $N_4 \leq 8$ and $N_3 \leq 20$ has been made in Ref. 146 within a VMC method, and in Refs. 147,152 within the DMC method. Systems with $N_4 \leq 17$ and $N_4 \leq 3$ have been considered in Refs. 153,154 within DMC. The stability of these systems is summarized in Fig. 11, where filled squares indicate stable systems, open squares indicate metastable systems, and no symbol corresponds to unstable systems. Four or more ^{4}He atoms guarantee the stability of any number of ^{3}He atoms. The first striking point of this Table is that a single ^{4}He atom is able to stabilize the otherwise unstable $^3{\rm He}_{20}$ system. As compared with the addition of a single ^{3}He atom, the only differences lie in the zero-point energy and the suppression of Pauli's principle, the interaction being the same. The second interesting point is that there are metastable systems, in which the DMC algorithm converged to a negative energy value, but higher than or equal to (within statistical errors) that of the cluster with one less fermion. These systems appear at the beginning and at the completion of the $1p$ and $1d$ shells, for $N_4 = 2$ and 3.

Concerning these metastable systems, the following comments are in order. First, an exact and unconstrained DMC calculation should converge to a bound subsystem with one less fermion plus a far away unbound fermion.

This is likely what would happen in simulations based in an approximate Green's function, if the total elapsed time were enormously long. However, the required projection on the importance sampling function, which is space confined, may prevent the time evolution to end up at the unbound system. Second, as the DMC calculation is based on a fixed-node approximation, it is not excluded that the metastable systems will be stabilized if this condition is released, at least for $N_4 = 3$.

The non-destructive transmission grating diffraction used in Refs. 50,131 has been employed[155] to observe the formation of mixed clusters with mass smaller than 31 a.m.u. The mass spectrometer was adjusted to detect the $(^3\mathrm{He}^4\mathrm{He})^-$ ion, thus ensuring that only mixed clusters are counted. Notice that in some cases a given value of the mass corresponds to two or three different clusters (N_4, N_3). In these cases, the theoretical results help in interpreting the experiment. There is an overall compatibility between the observed peaks and the stability chart of Table 11, with the only exception of the cluster (4,4) for which no peak is experimentally resolved.

3.3.3. Larger mixed droplets

When N_3 is increased, it turns out that the ^{3}He effective potential has a fairly large number of bound sp surface states. A ^{3}He layer develops at the surface of the $^4\mathrm{He}_{N_4}$ drop, forming a quasi two-dimensional spherical shell. This is illustrated in Fig. 12 for the two drops chosen in Ref. 156 as typical examples: (300, 50) and (1440, 288). In the lower panels are plotted the effective potentials V_4 and V_3. Both potentials are constant inside the system, and have a minimum at the surface, in the form of a slight dip for V_4 and an important minimum for V_3. The ^{3}He chemical potential intersects the potential well only at the surface. Thus, all ^{3}He atoms are pushed to occupy surface states, as is clearly reflected in the density profiles displayed in the upper panels of the same figure. This is always the pattern if N_3 is small enough as compared with N_4. The maximum number of ^{3}He atoms which can be accomodated in a single shell on the surface of a ^{4}He drop can be roughly estimated as $4\pi R^2 \Delta r \rho_3$, where $R \simeq 3\, N_4^{1/3}$Å is the radius of the ^{4}He drop, $\Delta r \simeq 2$Å is the 'size' of a ^{3}He atom, and $\rho_3 \simeq 0.016$ Å^{-3} is the bulk ^{3}He density. The surface of the drop can thus accommodate all ^{3}He atoms if the condition $N_3 \leq 3.5\, N_4^{2/3}$ is fulfilled.

Otherwise, when N_4 is much smaller than N_3, ^{3}He atoms have a sizeable probability of being dissolved in the bulk of the droplet. A discussion on how ^{3}He dissolves in ^{4}He droplets can be found in Ref. 157. The value $N_4 = 728$ was considered, because it is large enough to clearly distinguish surface and

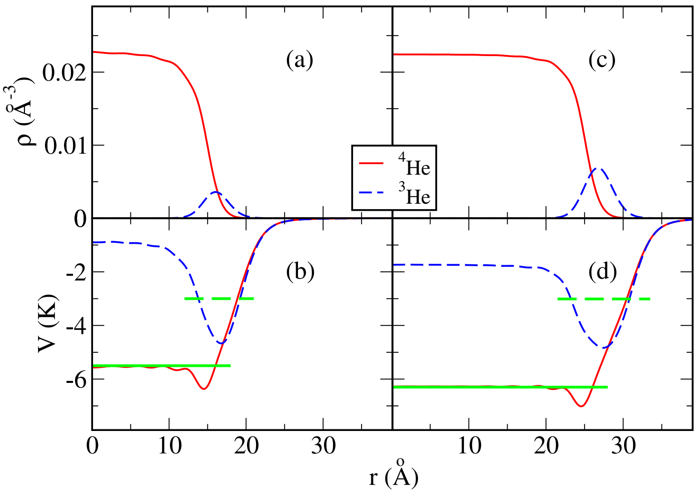

Fig. 12. (Color online) Effective sp potentials V_4 and V_3 (lower panels) and densities ρ_4 and ρ_3 (upper panels) for the droplets $^4He_{500}$+$^3He_{50}$ (left panels) and $^4He_{1440}$+$^3He_{288}$ (right panels). The horizontal dashed and solid lines in the lower panels represent the ^{3}He and ^{4}He chemical potentials respectively.

bulk regions in the corresponding pure drop. Figure 13 displays the situation in which a $^4He_{728}$ droplet is coated with an increasing number of ^{3}He atoms, and the limiting situation of the same drop immersed into liquid ^{3}He.

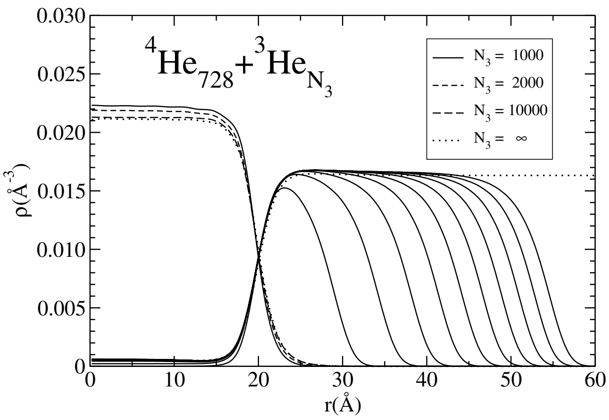

Fig. 13. Density profiles of $^4He_{728}$+$^3He_{N_3}$ droplets, for values of N_3 from 1000 to 10000, in steps of 1000.

It is apparent that a fairly large amount of ^{3}He is needed before it is appreciably dissolved in the interior of the ^{4}He part. The concentration of ^{3}He inside the ^{4}He$_{728}$ drop can be defined as the ratio $\rho_3(0)/[\rho_4(0) + \rho_3(0)]$ of densities at (or near) the origin. We recall that in the liquid mixture the solubility is 6.6% at zero temperature and pressure[18]. In finite systems, its value evolves from 6.25×10^{-5}% for $N_3 = 1000$ to 2.71% for $N_3 = 10^4$. The limiting solubility value (liquid ^{3}He) is 2.5%, slightly smaller than for large (728, N_3) droplets, which is an indication that finite size effects still appear in rather large drops. Another finite size effect is that the average ^{3}He density is above the saturation value even for the larger drops, showing that the existence of the outer ^{3}He surface still causes a visible density compression.

In systems $(N_4, 1)$ we have showed that when the ^{3}He atom is pushed to the surface region, the centrifugal term $\ell(\ell+1)/r^2$ entering the equation for the ^{3}He wave function can be treated as a perturbation, and consequently the ^{3}He spectrum corresponds to a series of rotational bands. The DF description of a (N_4, N_3) drop requires the self-consistent solution of a system of equations formally identical to Eq. (16), with the effective potentials V_4 and V_3 as well as the effective mass m_3^* depending on both densities ρ_4 and ρ_3. As far as the ^{3}He atoms are distributed in the surface region, a similar rotational spectrum should be expected. In Fig. 14 are displayed the sp energies as a function of $\ell(\ell+1)$ for the example cases of Ref. 156. They are distributed in two nearly parallel straight lines, one corresponding to the nodeless $n = 0$ states, and the other to the $n = 1$ states. The slope of these lines, as well as the gap between the last occupied $n = 0$ and the first unoccupied $n = 1$ sp state, diminish as N_4 increases.

We thus see that there are two energy scales clearly separated. The large one is related to the number of nodes of the radial wave function, and the small one to the different values of the orbital angular momentum for a given number of radial nodes. Therefore, when $N_4 \gg N_3$ the ^{3}He mean field gives rise to a distinct shell structure in which the sp energy levels group into rotational bands whose head states ϕ_{n0} are characterized by the number of nodes of their radial wave function. For drops satisfying the condition $N_3 \leq 3.5 N_4^{2/3}$, the Fermi level corresponds to a nodeless orbital, while the $n = 1$ sp states lie at higher energies. It can be seen in Fig. 14 that the $n = 1$ band starts overlapping the $n = 0$ band at $l_{cr} = 8$ for $N_4 = 300$, and at $l_{cr} = 14$ for $N_4 = 1440$. The total number of atoms that can be placed in the $n = 0$ bands up to l_{cr} is 162 and 450, respectively $[N_3 = 2\sum_0^{l_{cr}}(2l+1) = 2(l_{cr}+1)^2]$. These numbers are in very good agreement with the estimates given by the above inequality. We thus conclude that, as far as the inequality is respected, new magic numbers $N_3 = 2(p+1)^2$ appear, with $p = 0, 1, \ldots, l_{cr}$.

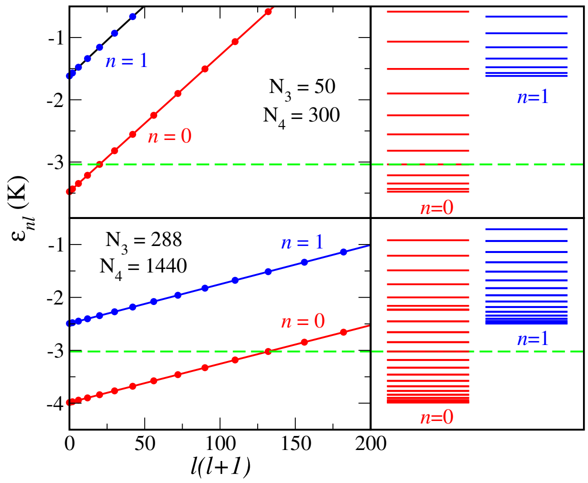

Fig. 14. (Color online) ^{3}He sp energies $\varepsilon_{n\ell}$ as a function of $\ell(\ell+1)$. The dashed horizontal lines represent the ^{3}He chemical potential.

4. DOPED HELIUM DROPLETS

In this Section we limit ourselves to the case of attractive impurities, and leave aside the case of electrons in helium clusters, referring the interested reader to the specific references given in the Introduction.

While doped ^{4}He nanoclusters where experimentally produced since the early 1990's, ^{3}He ones were first reported in 1997 (Ref. 158). Recent reviews on the current experimental situation can be found in Refs. 4,5,7. Reference 14 contains an updated revision of results obtained employing QMC techniques.

Helium clusters provide unique laboratories to test the onset and robustness of superfluidity in finite systems[38,39,159], which can be examined in terms of the response to an external perturbation, like the rotation of an embedded dopant. In this respect, the observation of sharp lines in the infrared spectrum of carbonyl sulfide (OCS) in ^{4}He drops[37] was the first evidence of decoupling between molecular rotation and motion of the helium atoms. In general, high-resolution spectroscopy in the microwave and infrared domain, as well as electronic spectroscopy, using the impurity as a spectroscopic probe, has been a valuable instrument to identify many aspects of the structure of helium clusters, like the solid or liquid nature of the first few solvation shells, their superfluid character *vs.* their capacity to adiabatically follow the rotation of the dopant, and characteristics of the helium-dopant interaction, revealed for example in the location of the latter either in the bulk[160] or at the surface[161] of the drop.

Charge transfer between helium and the dopant plays an essential role in the process of its capture and localization[162]. Together with fragmentation of ionized clusters, they constitute an active field of research (see e.g., Refs. 163,164 for updated references).

4.1. Strongly attractive dopants

Atoms or molecules whose interaction with helium is strongly attractive are found to reside in the bulk of the droplet. Within DF theory, this is illustrated in Fig. 15, where we show the density profiles of pure $^3He_{526}$ and $^4He_{526}$, and the same drops doped with a SF_6 molecule that interacts with helium through the spherically averaged pair potential of Ref. 165. In most DF, Hartree-like[166] and HNC/VMC approaches, massive impurities such as rare gases, SF_6 and various organic molecules, are regarded as fixed point sources at the center of the cluster and provide a helium-impurity potential $U(r)$ that enters the one-body mean field. DMC and PIMC calculations usually do not resort to this approximation, see e.g., Refs. 167,168. A comparison between the SF_6@$^4He_{111}$ densities obtained with PIMC (Ref. 122) and the OP functional can be found in Ref. 169, showing a rather good agreement.

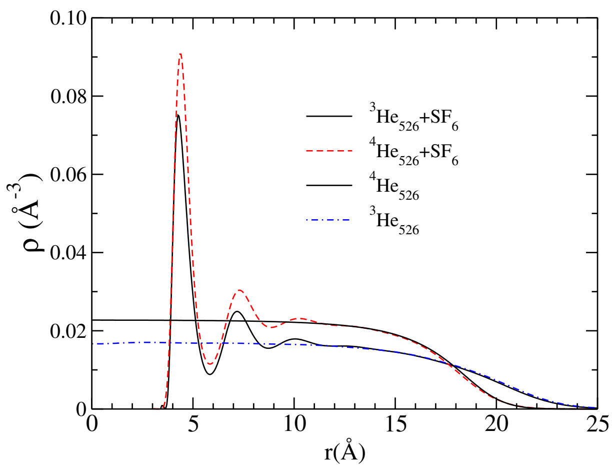

Fig. 15. (Color online) Density profiles of pure $^3He_{526}$ and $^4He_{526}$ drops, and of the same drops doped with SF_6.

Depending on the strength of the helium-dopant interaction, the first shell of helium atoms around the impurity is more or less compressed, remaining liquid[170] or even becoming solid -snowball[79,171]. This is a local effect that dies out at relatively short distances from the impurity, and hardly more than 2 or 3 solvation shells are clearly visible in the calculations. The number of He atoms in the shells also depend on the strength of the interaction. For instance, in the case of $SF_6@^4He_{40}$, DMC calculations yield about 23 ^{4}He atoms in the first solvation shell[167], in agreement with DF calculations[172]. In the latter reference, one may find the dependence of the occupation of the first solvation shell on the number of atoms in the droplet for some rare gas dopants and SF_6. As a general trend, the occupation for ^{3}He drops is about 5 atoms smaller than for ^{4}He drops. This is due to the combined effect of statistics and the lighter mass of the ^{3}He isotope.

A relevant quantity is the solvation energy for a generic dopant X defined as

$$S_\alpha(X) = E\left({}^\alpha\mathrm{He}_N + X\right) - E\left({}^\alpha\mathrm{He}_N\right) \ , \tag{17}$$

where E is the total energy of the pure or doped drop. Values of the solvation energy for rather attractive impurities obtained within DF theory[172] are shown in Fig. 16 as a function of the number of atoms in the droplet. As N grows for a given isotope, there is a well defined asymptotic trend towards the solvation energy for the selected impurity in bulk helium[169]. Statistics manifests in the fact that $S_3(X)$ is below $S_4(X)$, due to the smaller amount of fermion atoms in the first solvation shell (see Fig. 15), together with their enhanced kinetic energy[135].

As in pure helium droplets, an essential difference between ${}^3\mathrm{He}_N + X$ and ${}^4\mathrm{He}_N + X$ configurations is that the sp spectrum of the former possesses a shell structure with shell closures at well-defined magic numbers. We have seen in Sec. 3 that these numbers are quite different in pure ^{3}He and in mixed helium droplets. In doped ^{3}He drops these numbers also change, but not as dramatically. Another feature of ^{3}He drops doped with attractive impurities, like rare gases, is that within DF theory, ${}^3\mathrm{He}_N + X$ systems with $N \geq 8$, which is the smallest N used in the calculations[135], are bound. In this context, X plays the same role as ^{4}He atoms in mixed droplets (Sec. 3).

Microscopic studies of density distributions and energetic trends of small to medium size doped ^{4}He nanodroplets have been made employing the variational HNC/EL approach[123] and QMC (see e.g., the reviews in Refs. 11,14). In recent years, QMC has proven to be a substantial tool to investigate theoretically the coupling between the dopant rotation and the helium environment in small ${}^4\mathrm{He}_N$ clusters, together with the spatial localization of the superfluid fraction. In particular, PIMC investigations[171] of the microscopic structure of $\mathrm{Na}^+@{}^4\mathrm{He}_N$ up to $N = 100$ revealed that the first solvation

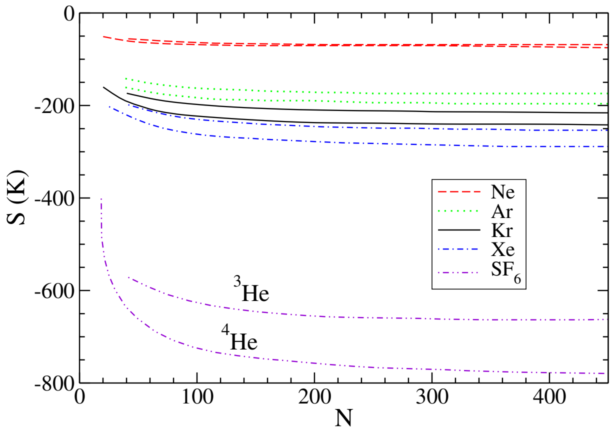

Fig. 16. (Color online) Solvation energy as a function of the number of He atoms in the drop. For each impurity, the upper and lower curves respectively correspond to ^{3}He and to ^{4}He.

shell is solid-like and that a superfluid transition takes place in the second shell at T between 1 and 1.25 K. More recent calculations for less attractive impurities have shown that for linear dopants[173], the above shell exhibits a decreased superfluid fraction. DMC calculations[174] of rotational states of SF_6 in ^4He_N, in quantitative agreement with experimental evidence[160], indicated that the foreign molecule behaves as a free rotor, with an enhanced moment of inertia. This increase ΔI originates in the fact that part of the helium density follows the rotation adiabatically. DF calculations based on the adiabatic following approximation showed good agreement between computed and observed ΔI for heavy rotors[175], and later QMC studies of structure and rotational dynamics involving other molecular rotors[176–179] reinforced the image that the rotation influences the microscopic structure of the first solvation shell, and that the lighter the impurity, the larger the number of ^{4}He atoms needed in order to saturate the rotational constant of the complex. In Ref. 178, the projection operator imaginary time spectral evolution (POITSE) method[180] was applied to compute rotationally excited states of HCN in ^4He_N. The results indicate that the low-lying rotational states are not strictly those of a free rotator. A microscopic two-fluid model, valid for heavy rotors[181,182], permits to extract the rotational constant B of the molecule, defined from its spectrum as $E_J = B\,J(J+1)$, employing the PIMC helium density as an input[183].

Recent DMC investigations of the effects of rotations of molecules of various sizes and symmetries such as SF_6, OCS, C_6H_6 and HCN, that make use of the intrinsic anisotropy of the molecule-helium interaction, permitted to elucidate further the adiabatic following of the helium environment, giving rise to the concept of 'incomplete adiabatic following'[184]. For the four dopants, molecular rotation affects significantly the solvating helium density, however the details of the respective density responses are different and reveal some trends with molecular mass and symmetry. The analysis of the helium densities shows that for the heaviest rotors, only a fraction of the density adiabatically follows the molecular motion in the gs, and this fraction is negligible for a light impurity like HCN. So, adiabatic following is never complete. As summarized in Ref. 185, these studies lead to the following conclusions: a) the rotational constant of heavy rotors is reduced by a factor 2-3 upon solvation, whereas the reduction is just of a few tens per cent in the case of light rotors; b) the rotational constant of solvated heavy rotors would reach the nanodroplet regime corresponding to e.g., the experimental conditions of Ref. 37, well before the first solvation shell is completed, whereas the convergence for light rotors would be much slower. These facts were understood on the concept of adiabatic following: heavy molecules rotate slowly and have more asymmetric potentials with He. Thus, some helium would be dragged along with the molecule, decreasing B. By contrast, light molecules rotate faster and have more spherical potentials, dragging along less helium.

Newer accurate simulations[185,186] that employ a QMC technique especially designed to address the dynamical properties of interacting boson systems, the so-called Reptation QMC (RQMC)[187], revise the different behaviors of light and heavy rotors with increasing host size, pointing out that rather than the molecular weight, it is the anisotropy of the molecule-He potential which determines the reduction of B upon solvation and the way it approaches the nanodroplet regime value. In these recent papers, application of RQMC to the calculation of the rotational constant of several cromophore molecules embedded in helium clusters reverse the previous scenario, since the resulting convergence of B to the asymptotic value is slow (fast) for heavy (light) dopants.

The above two different QMC methods illustrate the transition from a weakly bound molecular complex to a quantum solvated molecule[185,186,188–191], as shown by high resolution microwave and infrared spectroscopy[192–194]. Application of the POITSE method[188] to the rotational motion of an OCS molecule inside small helium clusters allowed for the first time to identify the above transition as a function of cluster size N. A similar trend was observed by applying RQMC to CO_2, CO and N_2O impurities (Refs. 189–191, respectively); B decreases as a function of N, reaches a minimum for cluster sizes between 5 and 9 and appear to converge towards the nanoscopic limit for N around 30 to 50. This is elloquently illustrated in Figs. 17 and 18,

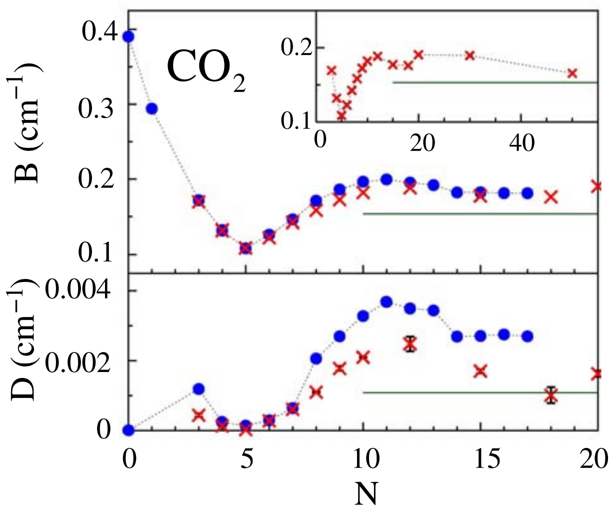

Fig. 17. (Color online) Rotational constants B and D defined from the rotational spectrum of CO_2 in ^4He_N as $E_J = B\,J(J+1) - D\,J^2(J+1)^2$. Circles and crosses indicate, respectively, the experimental and theoretical results[189]. The horizontal lines represent the nanodroplet result[195] (by courtesy of S. Moroni).

that represent RQMC results together with experimental data. In Fig. 17, the B and D rotational constants of CO_2 in ^4He_N from Ref. 189 are shown as a function of cluster size N, while in Fig. 18 we can visualize B for OCS in ^4He_N as reported in Ref. 185.

The appearance of the minimum at given N_m has been interpreted as the signal of the transition between the molecular complex that characterizes the configuration X@^4He_N for the lowest N's, to the quantum solvated system for N above N_m; this is a pure quantum effect traceable to the boson exchange symmetry that decreases the effective inertia of the first solvation shell, and is absent in PIMC calculations for Boltzmann's statistics[191]. Earlier RQMC simulations of OCS@^4He_N predict rotational constants in excellent agreement with experiment for $N \leq 5$ (Ref. 196), but overestimate the values for higher N's up to 20 atoms. Improved helium-impurity potentials for OCS and N_2O were employed in Ref. 197, together with the POITSE method, yielding a comparable good fit to experimental figures for low N.

A recent theoretical study that employs a combined HNC/EL+DMC technique[198] sheds light on the changes in the rotational spectrum of a light rotor due to coupling to the phonon-maxon-roton excitations of the helium solvent; experimental evidence of such an effect in solvated ethylene molecules in ^4He clusters has been pointed out recently[199]. A newest appli-

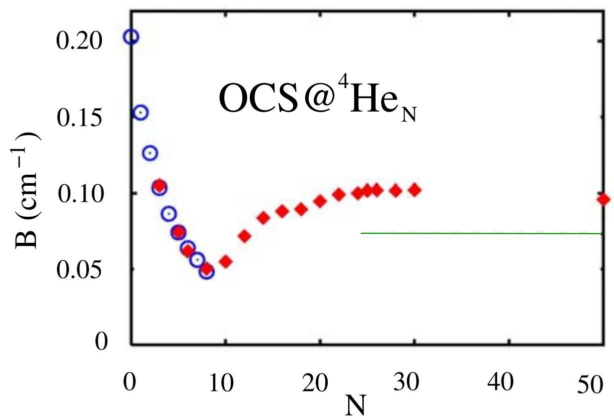

Fig. 18. (Color online) Rotational constant B of OCS in ^{4}He$_N$. Circles and diamonds indicate, respectively, the experimental[192] and theoretical[185] results. The horizontal lines represent the nanodroplet results[192] (by courtesy of S. Baroni and S. Moroni).

cation of the PIMC-POITSE method[200] showed theoretical evidence for the onset of superfluidity in small CO_2@^{4}He$_N$ clusters in the turnaround of the rotational constant at $N \sim 5$ atoms.

In the current perspective, one appealing situation is that of isotopically mixed, doped clusters, first addressed theoretically in Ref. 112. Experimental studies of large ^{3}He drops with low admixture of ^{4}He and capture of SF_6 indicate that fermion atoms rearrange in a shell around a ^{4}He core that hosts the molecule in its bulk[17,160]. Furthermore, the Andronikashvili-like experiment carried by Grebenev *et al*[37] shows that a system with around 60 ^{4}He atoms is superfluid, as demonstrated by the free rotation of an OCS molecule trapped in a mixed cluster. This is illustrated in Fig. 19, and it is in agreement with earlier theoretical predictions[38,39] (see however Ref. 201 for another possible interpretation of the experiment). Later on, Toennies and his group have used the same strategy to show that para-hydrogen (pH_2) clusters are superfluid[202], analyzing the infrared spectrum of OCS inside a pH_2 cluster embedded into a mixed ^{4}He-^{3}He drop.

By contrast, ^{3}He atoms remain normal in both pure and mixed drops. A systematic study of the evolution of large mixed clusters with large fermion numbers was thus called for. Within DF theory, one such study has been carried out in Ref. 157. The role of boson-fermion statistics in the rovibrational spectrum of a Br_2 molecule in a ^{3}He$_{18}$ drop with $N_4 = 0, 6$, and 18 ^{4}He atoms added to it has been recently investigated[203].

Fig. 20 shows density profiles for ^{4}He$_{728}$+^{3}He$_{N_3}$+SF_6 droplets and sev-

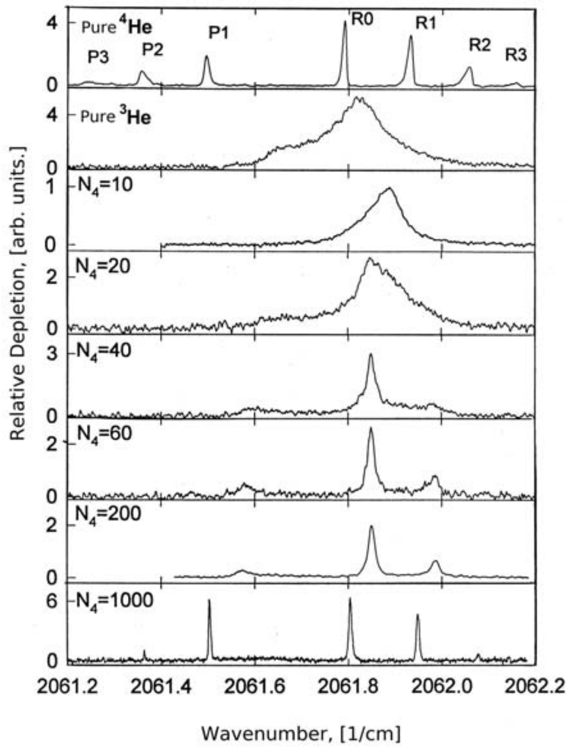

Fig. 19. Rovibrational absorption spectrum of a OCS molecule in helium droplets. The top panel corresponds to a $^4He_{6000}$ drop, and the next one to a $^3He_{12000}$ drop. The other panels correspond to cases in which a progressive number of ^{4}He atoms N_4 has been added to the pure ^{3}He drop. The RJ peaks correspond to $J \rightarrow J+1$ transitions, and the PJ ones to $J \rightarrow J-1$ transitions, J being the angular momentum of the molecule. The neat peaks observed in the pure ^{4}He drop mean that the OCS freely rotates in a superfluid environment. The appearance of well defined peaks at about $N_4 = 60$ is interpreted as a signature of the onset of superfluid manifestations (by courtesy of J.P. Toennies).

eral values of N_3 (Ref. 157). It shows that ^{3}He remains at the surface of the droplet even for large N_3 values -the small peak at $\simeq 7$ Å in the ^{3}He density profile corresponds just to about one single fermion atom.

More relevant in the context of the experiment discussed in Ref. 37 are the density profiles displayed in Fig. 21, and complementary results given in Ref. 157. This figure shows that when the number of ^{4}He atoms exceeds the maximum population of the first solvation shell, depending on the N_3/N_4

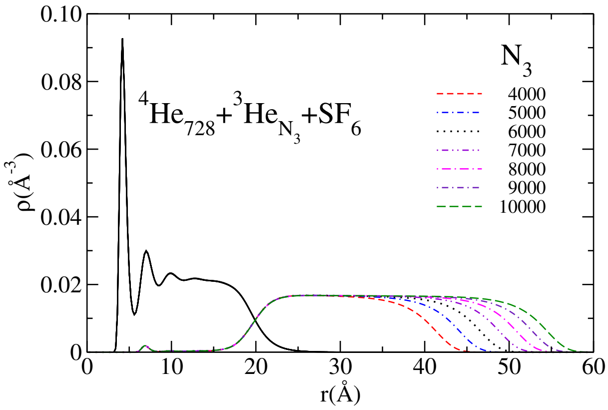

Fig. 20. (Color online) Density profiles of $^4He_{728}$+$^3He_{N_3}$+SF_6 droplets for several values of N_3.

ratio, the second solvation shell may contain an appreciable number of ^{3}He atoms. This is the scenario discussed in Ref. 37, and the calculations thus support the notion that the impurity remains in a superfluid environment.

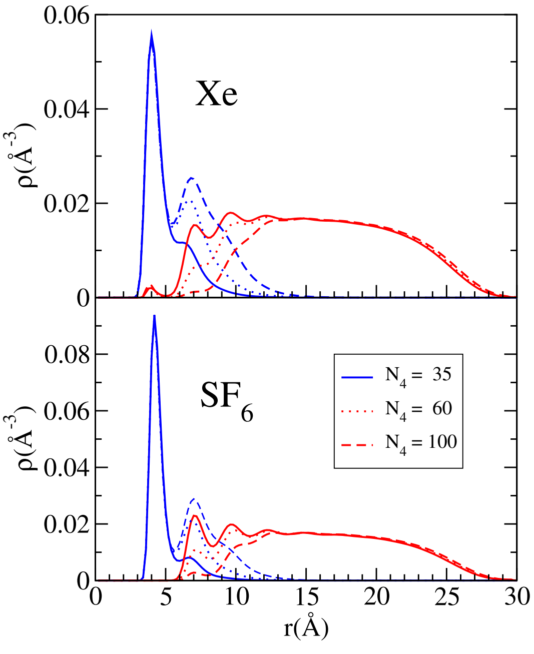

Fig. 21. (Color online) Density profiles of $^4He_{N_4}$+$^3He_{1000}$+SF_6 for $N_4 = 35$, 60, and 100 (bottom panel), and the same for Xe (top panel).

4.2. Weakly attractive dopants

If the dopant-helium interaction is weakly attractive, the impurity seeks a gs configuration near or at the surface of the cluster, as is the case for alkali (Ak)[34,161,204,205] and, to some extent, for alkaline earth atoms[206,207]. This has been experimentally established by laser-induced fluorescence (LIF) and beam depletion (BD) spectroscopy[5], as evidenced by the much narrower and less-shifted spectra when compared to those found in bulk liquid ^{4}He[34,161,204–206]. For alkalis, this result was predicted by DF[208] and confirmed by PIMC[168] calculations, both yielding alkali binding energies of a few Kelvin, in agreement with the measurements of detachment energy thresholds using the free atomic emissions[209].

Fig. 22 shows the density profiles of ^{3}He$_{2000}$ and ^{4}He$_{2000}$ drops doped with an alkali atom[210,211] (the case of Ca@^{4}He$_{300}$ is shown in Sec. 6). It can be seen that, contrarily to the case of strongly attractive impurities, the helium density is only slightly perturbed by the presence of the impurity, which produces a 'dimple' on the droplet surface. Comparison with the stable state on the ^{4}He cluster shows that the 'dimple' structure is more pronounced in the case of ^{3}He, and that the alkali impurity lies *inside* the surface region for ^{3}He and *outside* the surface region for ^{4}He (the surface region is usually defined as that comprised between the radii at which $\rho = 0.1\rho_0$ and $\rho = 0.9\rho_0$). This is due to the lower surface tension of ^{3}He as compared to that of ^{4}He, which also makes the surface thickness of bulk liquid and droplets wider for ^{3}He than for ^{4}He[47,48].

The deformation of the surface upon alkali adsorption is characterized by the 'dimple' depth, ξ, defined as the difference between the position of the dividing surface at $\rho = \rho_0/2$, with and without impurity, respectively[208]. For ^{3}He$_{2000}$, the DF values are ξ = 4.3, 4.1, 4.3, and 4.6 Å for Na, K, Rb and Cs, respectively. The corresponding values for ^{4}He$_{2000}$ are ξ = 2.2, 2.2, 2.3, and 3.3 Å, respectively. DF calculations thus confirm the picture emerging from the measurements[211], i.e. (i) the surface location of Na on ^{3}He nanodroplets, and (ii) a more pronounced 'dimple' than in ^{4}He droplets.

Knowledge of the detailed structure of the 'dimple' allows to calculate the shift between the Na@^{3}He$_N$ and Na@^{4}He$_N$ BD spectra within the Franck-Condon approximation, i.e. assuming that the 'dimple' shape does not change during the Na excitation. It has been found that the ^{3}He spectrum is blue-shifted with respect to the ^{4}He one by 6.4 cm^{-1}, in good agreement with the experimental value of 7.5 ± 1 cm^{-1} (see Fig. 23)[210,211].

These results show that alkali adsorption on ^{3}He droplets occurs in very much the same way as in the case of ^{4}He. The similarities in the experimental spectra are certainly remarkable for two apparently very different fluids, one

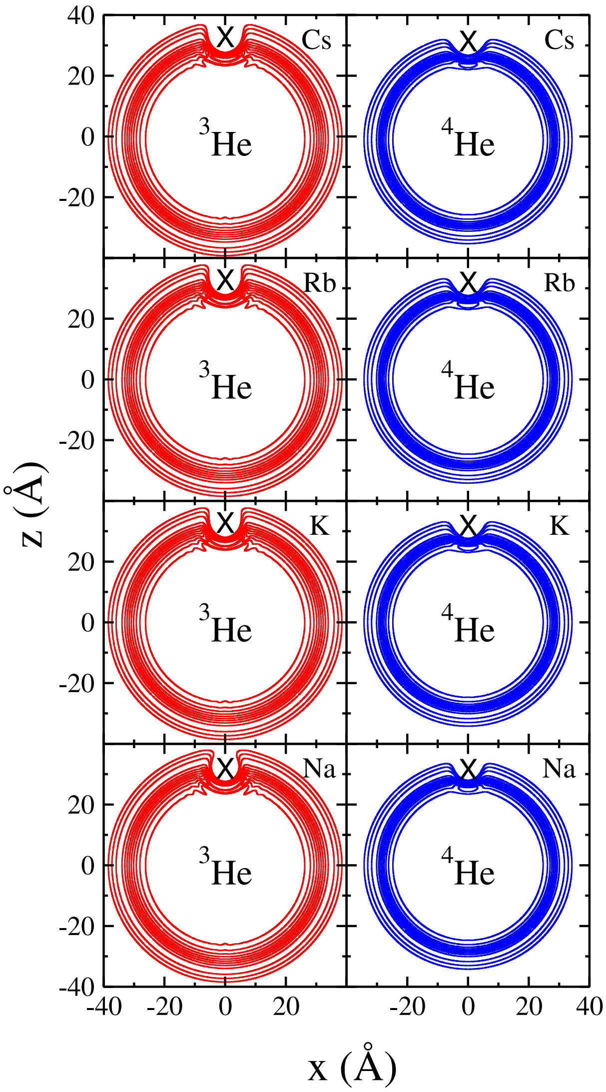

Fig. 22. Equidensity lines showing the stable state of an alkali atom (cross) on a He_{2000} droplet. The 9 inner lines correspond to densities 0.9 ρ_0 to 0.1ρ_0, and the 3 outer lines to $10^{-2}\rho_0$, $10^{-3}\rho_0$, and $10^{-4}\rho_0$ (ρ_0 = 0.0163 Å^{-3} for ^{3}He, and 0.0218 Å^{-3} for ^{4}He).

normal and the other superfluid, and clearly indicate that superfluidity does not play any substantial role in the processes. This is likely a consequence of the very fast time scale characterizing the Na electronic excitation compared to that required by the He fluid to readjust. The excitation thus occurs in a 'frozen' environment and the only significant difference between ^{3}He and

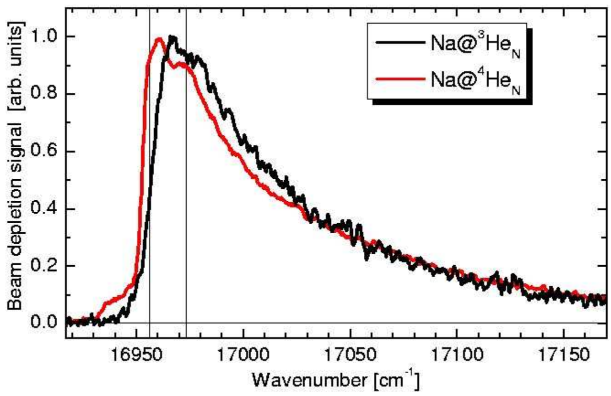

Fig. 23. (Color online) Beam depletion spectra of Na atoms attached to ^{3}He/^{4}He droplets with about 5000 atoms. The vertical lines indicate the positions of the two fine structure components of the Na gas-phase $3p \leftarrow 3s$ transition. (by courtesy of F. Stienkemeier).

^{4}He is due to the different 'dimple' structure, which accounts for the small shift in their spectra observed in the experiments.

'Solvation energies' (actually, alkalis are not solvated, but Eq. (17) still makes sense, and $|S_\alpha(Ak)|$ represents the binding energy of the attached alkali atom) have been also calculated and are shown in Fig. 24. The value $S_4(Na) \sim -12$ K has been obtained within DF theory for Na adsorbed on the planar surface of ^{4}He, which corresponds to the $N = \infty$ limit[208]. In contrast with the rather attractive impurities displayed in Fig. 16, the alkali 'solvation energies' do not show an intuitively simple trend. In particular, heavy alkalis (K, Rb) are more bound in ^{4}He drops than in ^{3}He ones, opposite to the lighter Na.

An interesting process may take place in AkHe$_N$ drops when alkalis are excited on the surface of the drop. This is the formation of AkHe exciplexes, i.e., excited di- or multiatomic complexes that are not stable in the electronic gs of the Ak [AgHe$_2$ exciplexes were firstly observed in bulk liquid ^{4}He (Ref. 212)]. In the case of ^{4}He, they have been identified from their free fluorescence emission spectra for Na, K, and Rb in ^{4}He clusters [34,209,213,214]. The formation process of K^{4}He exciplexes has been investigated in real time by means of femtosecond pump-probe spectroscopy[215]. Recently, these experiments have been extended to ^{3}He drops doped with Rb, allowing a detailed comparison of the results obtained for both isotopes[216]. The theoretical interpretation of the formation times looks as quite a formidable task[217]. Simpler approaches, based on a semiclassical model of tunneling across the Ak-He-He$_N$ molecular potential, have been employed to address this prob-

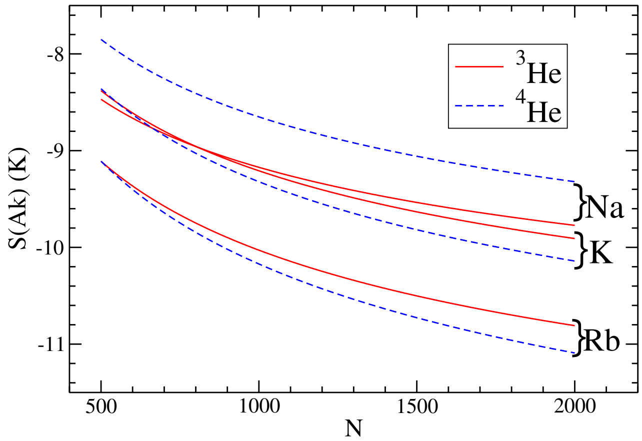

Fig. 24. (Color online) Alkali 'solvation energies' as a function of the number of He atoms in the droplet.

lem. This model has been found to work in some Ak^4He_N cases, but fails for Rb^3He_N drops[216], in spite of using as input the same basic ingredients. Although the model used for the theoretical analysis is too crude, this might indicate that a different relaxation dynamics of the superfluid is responsible for the observed formation times.

Dynamical DF studies might help to shed light on this problem, in which case a first step is to solve the static EL equations relaxing the approximation of considering the alkali as a fixed point source of a helium-impurity potential, i.e., one has to take into account the zero point motion of the alkali and solve the EL equation for the liquid coupled to a Schrödinger-like equation for the Ak atom. This is unavoidable in the case of the lightest alkali, Li. It is also relevant to know the binding energy of Li to ^3He_N droplets in view of current experiments[218], as well as the 'vibrational spectrum' of an alkali atom in the field created by the helium droplet. All these studies are very sensitive to the impurity-He pair potential, not always known with enough accuracy.

Fig. 25 shows the results for Li@He_{300} worked out using the Li-He potential of Ref. 219; similar qualitative results are obtained using the more accurate pair potential of Refs. 220,221. These results have been obtained by solving the coupled Euler-Lagrange and KS equations for He and Li, respectively, treating ^3He in the TF approximation [222]. The 'solvation energies'

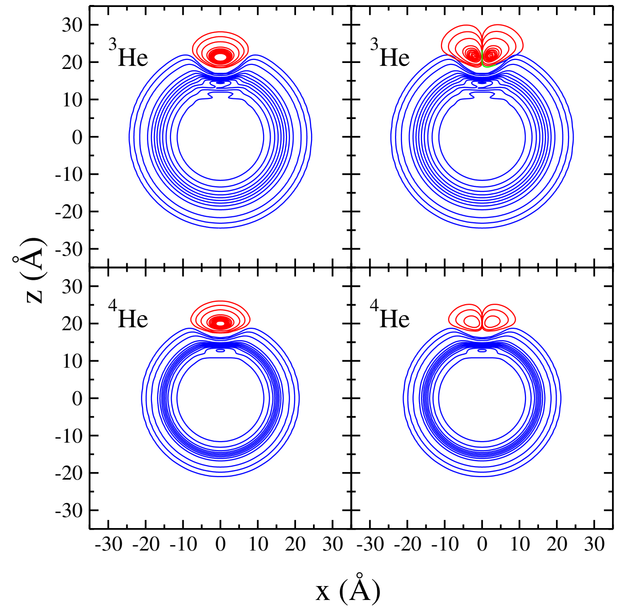

Fig. 25. (Color online) Left panels: equidensity lines showing the stable state of a Li atom on a He_{300} droplet. The 9 inner lines correspond to densities 0.9 ρ_0 to $0.1\rho_0$, and the 3 outer lines to $10^{-2}\rho_0$, $10^{-3}\rho_0$, and $10^{-4}\rho_0$ ($\rho_0 = 0.0163$ Å^{-3} for ^{3}He, and 0.0218 Å^{-3} for ^{4}He). Right panels: Same as left panels for the first excited state of a Li atom. The equidensity lines of the Li probability densities are shown in the same scale as the corresponding He density

S(Li) are -2.89 K for Li@^{3}He$_{300}$, and -3.91 K for Li@^{4}He$_{300}$. These values can be compared with the gs energies ϵ_{00}(Li) which are -2.45 K and -3.25 K, respectively. For either isotope, S(Li) and ϵ_{00}(Li) do not coincide. Actually, they should not, since $|S(\mathrm{Li})|$ and $|\epsilon_{00}(\mathrm{Li})|$ respectively represent the extraction energy of the Li atom in an adiabatic and in a sudden process.

Fig. 25 also shows that helium drops may host alkali atoms in excited states of the droplet mean field, even if the He-Ak potential is very shallow, especially in the direction perpendicular to the radius of the droplet. This potential it is very asymmetric in the z direction, as can be inferred from the Li probability densities shown in that figure. The first excited state of Li in Li@^{3}He$_{300}$ is at $\epsilon_{01} = -1.03$ K, and at -2.01 K in Li@^{4}He$_{300}$. These states have been obtained keeping frozen the helium density corresponding to the configurations displayed in the left hand side of that figure, which is justified for the lighter alkalis. Clearly, the DF Li spectrum is more realistic than that of crude one-dimensional models often used to address exciplex formation (see e.g. Ref. 216 and Refs. therein), as it takes into account the three dimensional structure of the He_N-Ak potential through a detailed description of the 'dimple'.

5. DYNAMICAL PROPERTIES OF HELIUM DROPLETS

5.1. Excited states and linear response

There is no direct experimental information on the spectrum of He droplets. LIF spectra of dopant molecules in large ($N \geq 10^3$) ^{4}He clusters are characterized by a sharp zero-phonon line, together with a broad phonon wing sideband which reflects the coupling of the molecular degrees of freedom with the collective phonon-roton modes of the droplet[223]. In the case of ^{4}He drops, the zero-phonon line and the phonon wing parts of the spectrum are separated by a gap, which is absent in the case of ^{3}He drops because of the particle-hole excitations characteristics of normal ^{3}He (Ref. 224). The interpretation of the experimental phonon wing structure in $C_2H_2O_2$ (glyoxal) is an indirect evidence of the phonon-roton modes in large ^{4}He doped drops. Diffraction of small, pure ^{4}He clusters has recently provided another indirect –but far more quantitative– evidence of collective modes of ^{4}He droplets[50].

5.1.1. Pure helium clusters

The first calculations of collective excitations in pure ^{4}He clusters were reported in 1990[39,113,225]. The analysis of the compressional modes of ^{4}He$_N$ clusters[39] pointed out that the spectrum of ^{4}He$_{70}$ displayed some of the characteristics of roton excitations in bulk liquid ^{4}He. This indicated that this cluster was superfluid, in agreement with PIMC calculations carried out almost simultaneously[38].

We collect in Table 2 collective energies and chemical potentials of selected ^{4}He$_N$ drops calculated within DF theory[115]. Similar results have been obtained by other authors[13,94,113,114,123], so that the emerging physical picture is robust.

Some aspects of the multipole spectrum of pure ^{4}He$_N$ drops make these systems rather distinct. On the one hand, their lowest lying surface -ripplon-mode with $L = 2, 3\dots$ exhausts the energy-weighted sum-rule (EWSR). This means that there is only one collective state carrying appreciable strength for each value of L. On the other hand, except for very small droplets, these states are in the discrete region of the energy spectrum, i.e., their excitation energy is smaller than the threshold energy for atom emission,

Table 2. Collective energies (K) and chemical potential μ (K) of selected $^4\mathrm{He}_N$ drops.

N	μ	$L=0$				$L=2$	$L=3$
		$\hbar\omega_1$	% EWSR	$\hbar\omega_2$	% EWSR		
40	-3.54	3.04	73	5.32	25	1.35	2.19
70	-4.16	3.22	73	5.35	23	1.11	1.96
112	-4.40	3.27	73	5.37	24	0.89	1.62
240	-5.00	3.16	82	4.77	16	0.62	1.19
728	-5.61	2.69	95	4.33	3	0.35	0.71

$|\mu|$. As a consequence, these states are weakly damped, with fairly large mean lives. This has been crucial for the interpretation of the experimental size distribution of small neutral ^{4}He clusters[50], as discussed below.

The compressional $L=0$ mode is fragmented, with more than one peak carrying sizeable strength. The evolution with N of the compressional mode displayed in Table 2 is quite general, holding also for surface modes: as N increases, the location of the mode evolves from the continuum –above atom emission threshold– to the discrete energy region of the spectrum. An updated discussion of the monopole mode can be found in Ref. 226. The response of droplets to a plane wave operator aiming at studying the phonon-roton curve in ^{4}He droplets has been addressed microscopically in Refs. 13,94,123, and in Ref. 114 using the Orsay-Trento DF, that reproduces by construction the dispersion of the elementary excitations in bulk liquid ^{4}He.

The size distribution of ^{4}He clusters in cryogenic jet beams was analyzed in Ref. 50 by diffraction from a transmission grating[129] for $N \leq 100$. Distinct peaks were observed in the cluster distribution at 'magic numbers' $N = 10-11, 14, 22, 26-27$ and 44 atoms. These numbers are explained by analyzing the growth of clusters in the early stages of the expansion under near-equilibrium conditions. The calculation of the equilibrium constants of growth reaction requires for each size N the detailed knowledge of the chemical potential and energies of excited levels characterized by (n, ℓ), which are respectively the radial and the angular momentum quantum numbers. These energies were systematically computed[50,126] by the DMC method for each N between 3 and 50. The partition functions calculated from these excitation levels increase smoothly with N, except when a new level is bound giving rise to a sudden step. The calculations indicate that levels (0,2) and (1,0) become stabilized at $N = 8 \pm 1$, (0,3) at 14 ± 1, (0,4) at 25 ± 1, and (0,5) at 41 ± 1. These numbers are in nice agreement with the experimental magic sizes, except for the enhancement at $N = 22$. In contrast with the situation in atoms, metallic clusters and atomic nuclei, these magic sizes are not related to enhanced binding energies at specific values of N where fermionic shells are closed. Instead, they are related to threshold sizes for which the quantized excitations are stabilized. This analysis gives thus indirect experimental evidence on excited states of ^{4}He droplets.

Systematic calculations of the collective spectrum of ^{3}He droplets have been only carried out in the RPA within DF theory[116,227,228] for magic clusters up to $N = 112$, and for larger clusters using the sum-rules or fluiddynamical approaches[105,229]. As a general rule, it has been found that both the surface and volume modes are more fragmented for ^{3}He than for ^{4}He clusters, especially for $L \geq 3$ (Ref. 227). For ^{3}He$_N$ clusters it has been also found that as N increases, the energies of the collective modes evolve from the continuum to the discrete region of the spectrum. We want to stress that even the largest ^{3}He$_N$ addressed in these studies, is much smaller than the smallest *pure* ^{3}He droplet ever detected experimentally, which has a few thousand atoms.

To the best of our knowledge, no systematic study of the collective modes of mixed helium drops exists, in spite that fairly small mixed ^{3}He-^{4}He droplets have been detected experimentally, and their spectrum could be theoretically addressed, at least within DF theory.

5.1.2. Doped helium clusters

The response of doped helium clusters has also been addressed in detail for ^{4}He[115,123,230], and more recently, for ^{3}He clusters[117,172] as well. In all these calculations, the impurity was treated as a point-like particle residing at the center of the drop –only fairly attractive impurities were considered in these studies–, and its zero-point motion was neglected. Since the effect of the impurity is always local, as we have discussed in Sec. 4, it is clear that one should not expect any sizeable change in the collective spectrum of large droplets, and thus the theoretical effort has been concentrated on small and medium size ones. For both isotopes, the outcome of the calculations is that the presence of impurities causes an appreciable fragmentation of the collective spectrum of these clusters, especially the volume monopole mode, as expected. Results for this excitation in several ^{4}He$_N$ drops doped with Xe and SF_6 are collected in Table 3 (Ref. 230), and can be compared with those of Table 2. Surface modes with $L = 2, 3 \ldots$ are less affected by the presence of an impurity in the bulk of the droplet.

A distinct feature of the collective spectrum of doped droplets is the

Table 3. Energies (K) of the collective monopole states, and their contribution to the EWSR for ^4He_N clusters with Xe and SF_6 impurities.

N	Xe		SF_6	
	$\hbar\omega_0$	% EWSR	$\hbar\omega_0$	% EWSR
40	4.33	46	5.76	41
	5.83	15	7.78	16
	7.48	14	10.01	23
70	3.86	64	4.41	56
	5.67	17	6.27	14
	7.55	13	8.19	12
112	3.66	65	3.94	59
	5.45	17	5.75	17
	7.03	12	7.38	13
240	3.41	80	3.47	75
	5.26	13	5.36	15
	5.47	5	6.60	7
728	2.81	94	2.89	94
	4.66	5	4.78	5
	7.37	1		

existence of a dipole $L = 1$ mode. This mode is absent in pure helium clusters because it would correspond to a translation of the droplet as a whole. The same happens for the atomic nucleus, that lacks 'isoscalar' dipole excitations, at variance with i.e., the case of the Sun, for which the dipole mode is a true excitation mode because of the large compressibility of its constituent gas. In the scenario described above, the dipole mode arises from the oscillation of the droplet as a whole in the 'external field' created by the impurity, and it is similar to the dipole -plasmon- mode in simple metal clusters, and to the magnetoplasmon mode in quantum dots[231].

As an illustrative example of the response of a ^{4}He cluster doped with a strongly attractive bulky impurity[232], we show in Fig. 26 the strength function of C_{120}@$^4He_{500}$. All the peaks displayed in the figure are below the atom emission threshold, which is at 7.5 K for this cluster.

Results for the $L = 0, 1$ and 2 modes of Xe@^3He_N and SF_6@^3He_N clusters are shown in Fig. 27. The corresponding results for ^{4}He clusters can be found e.g. in Ref. 230. Note again that, as N increases, the modes move from the continuum into the discrete region of the energy spectrum.

The evolution of the bulk monopole $L = 0$ and $L \geq 2$ ripplon modes

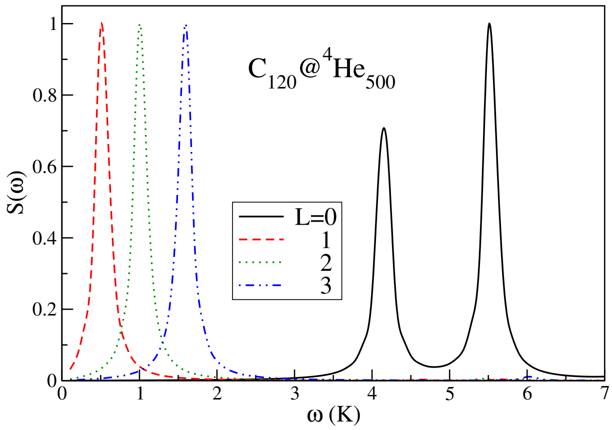

Fig. 26. (Color online) Strength functions for $C_{120}@^4He_{500}$. They have been normalized so that, for each L, the maximum peak height is unity.

for large droplets is easy to understand in the light of the LDM expressions of Sec. 2, which work well for large N's. The behaviour of the $L = 1$ mode deserves a comment. It is obvious that for rather attractive impurities –those residing in the bulk– and fairly large drops, a dipole displacement eventually corresponds to a simple translation of the impurity in the bulk of the drop. This is not an intrinsic excitation of the system and, as a consequence, the energy of the mode must go smoothly to zero as N increases. In contrast, for impurities residing on the surface of the droplet, the dipole mode persists even in the macroscopic limit –liquid free surface.

¿From the theory viewpoint, the fact that small helium droplets may sustain *stable* dipole excitations when doped with attractive impurities purposely located in its center, means that this is a stable configuration, and that these dopants do reside in the bulk of the droplet. Nowadays, this is an irrefutable experimental fact, but it the past it was a subject of debate among theorists[115,123,167,233].

Clearly, the spectra above described are very simple, as the dopant agent is spherically symmetric (rare gases), or its interaction with He has been spherically averaged (SF_6). Current interest has moved to the study of more complex impurities, such as e.g. planar aromatic molecules[234] doped with few ^4He atoms that do not completely solvate the impurity. In this case, one may address the evolution of helium excitations from localized, arising from few atoms located at the minima of the very anisotropic impurity-He potential, to collective excitations of the system[235].

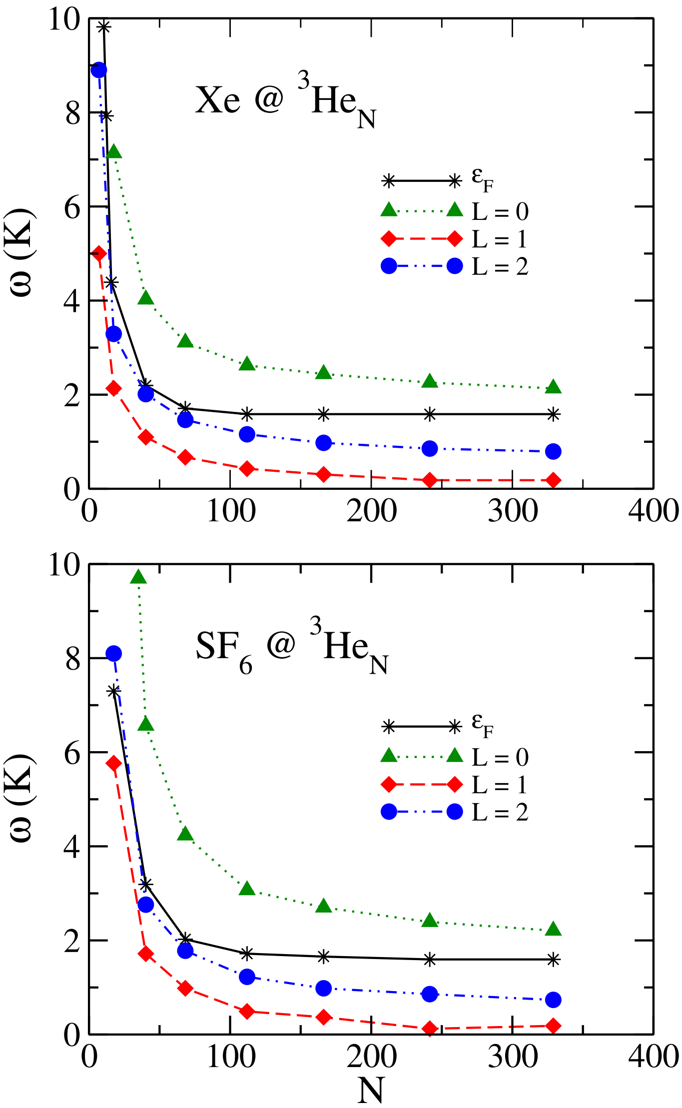

Fig. 27. (Color online) Mean excitation energies (K) and chemical potential in absolute value, ε_F (K), for ^3He_N droplets doped with SF_6 and Xe. The lines are drawn to guide the eye.

5.2. Large amplitude motions

For the sake of completeness, we close this Section commenting, in a non-exhaustive way, some processes that involve large amplitude motion of helium droplets. By far, the most relevant process of this kind, still an open question, is the actual dynamics of impurity capture by helium droplets, the subsequent energy and angular momentum deposition in the droplet, and their eventual dissipation by atom evaporation or even droplet fission. For ^{3}He clusters, these phenomena bear a strong similarity with deep inelastic and fission processes in heavy-ion nuclear reactions, that have been successfully described in the past using formalisms based on the Vlasov equation, when two-body collisions can be safely neglected, or on the Boltzmann-Uehling-Ulhenbeck and the Landau-Vlasov ones when two-body collisions are expected to play a key role[236,237]. These methods are appropriate for fermions, but cannot be applied to boson condensates, for which one possibility is to use TDDF theory, provided two-body -collisional- effects can be neglected.

^{3}He-^{3}He drop collisions in the Vlasov dynamics were described some time ago[238]. Since experimental data are not available, these calculations are predictive and show that the method can be implemented for these drops. With some affordable extra work, it could be extended to the more realistic Landau-Vlasov frame. Other aspects of the fission and fusion of ^{3}He droplets, and in particular the shape transitions as functions of the deposited angular momentum, have been also described[239].

These studies could be also extended to the ^{4}He case within TDDF theory, aiming at studying the evolution of an excited ^{4}He$_N$ cluster after the first stage of copious atom evaporation. As an illustrative example, we show in Fig. 28 the dynamical evolution of a ^{4}He$_{300}$ droplet obtained with the OP functional without the backflow term[240]. The starting configuration is the ground state boosted in the z direction with a large quadrupole energy, namely

$$\Psi(t=0,\mathbf{r}) = e^{\imath\lambda Q_2(\mathbf{r})}\sqrt{\rho(\mathbf{r})} \ , \tag{18}$$

where $Q_2(\mathbf{r}) = 2z^2 - x^2 - y^2$ is the quadrupole operator, and λ is a parameter whose value determines the energy initially deposited in the droplet. If λ is very small, the Fourier analysis of $Q_2(t)$ yields the energy of the $L = 2$ ripplon mode. In the present case, it has been chosen so that an energy of about 800 K is deposited into the droplet as a quadrupole velocity field, $\mathbf{u}(\mathbf{r}) = \lambda\hbar/m\nabla Q_2(\mathbf{r})$ The gs total energy of ^{4}He$_{300}$ is ~ -1383.6 K, and during the dynamical evolution, the energy of the system is conserved with high accuracy, $|\Delta E/E| \leq 5\times 10^{-4}$.

In spite of not being too realistic, as only mean field effects are taken into account and atom evaporation is hindered, this example shows that the method is workable even for fairly large excitation energies. It also shows that the fission process of a neutral system (see also Ref. 238) is rather different from that of a charged one, like atomic nuclei[241,242], or multiply charged simple metal clusters[243,244]. Indeed, the lack of a net electric charge makes

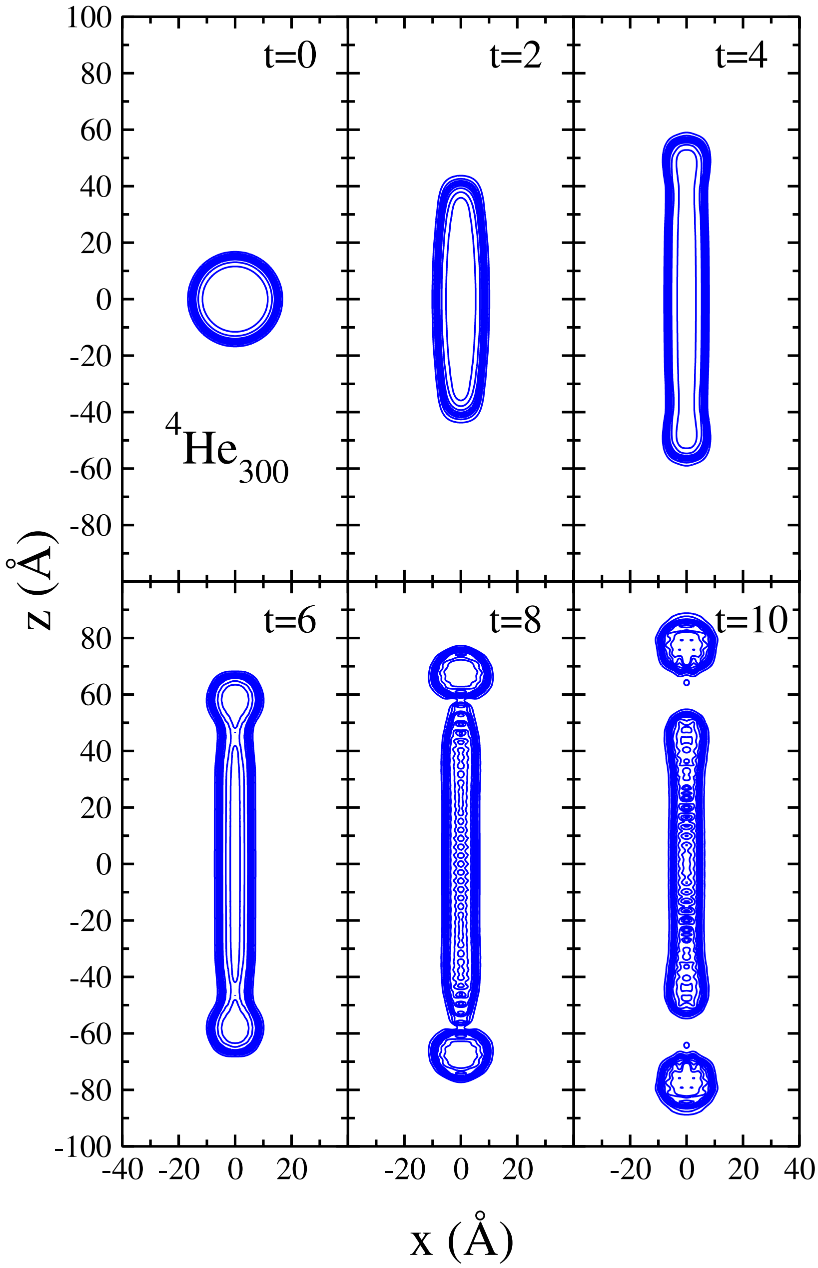

Fig. 28. (Color online) Equidensity lines showing the time evolution of a $^4He_{300}$ droplet to which a quadrupolar boost has been given. Time is shown in each panel in units of 7.64×10^{-12} sec. The lines correspond to densities $0.9\rho_0$ to $0.1\rho_0$ ($\rho_0 = 0.0218$ Å^{-3}).

the saddle fission configurations of helium droplets very different from the ones corresponding to these more 'conventional' systems. Ternary fission[242] or multifragmentation processes -notice the series of ridges eventually appearing along the neck, perpendicularly to the expansion direction- seem also likely to appear in these highly excited droplets.

Another relevant process involving large amplitude motions is that of molecular photodissociation in He clusters (see e.g. Ref. 245 for a review of the subject in other cluster environments), as theoretically addressed[246] using a hybrid model that takes into account the internal degrees of freedom of the dopant and describes the ^{4}He droplet within a path integral centroid molecular dynamics method[217,246]. Experiments on the translational dynamics of CF_3 in ^{4}He droplets by photodissociating CF_3I have been recently reported[247], whose theoretical interpretation would also require a full dynamical approach.

6. QUANTIZED VORTICES IN SUPERFLUID HELIUM NANODROPS

It has been recognized for a long time that any container of superfluid ^{4}He, treated in a conventional fashion, will be permeated *ab initio* by numerous quantized vortices stabilized by surface pinning, which generate self-sustained vortex tangles. We have discussed in previous sections that ^{4}He nanoclusters, as well as the bulk region of mixed helium droplets, are superfluid. It is thus natural to wonder about the appearance of quantized vortices in isolated droplets[248], since such structures have been created and detected in Bose-Einstein condensates made of dilute atomic gases. We refer the reader to Ref. 249 for a discussion of the analogies and differences between these two quantum systems.

The possible mechanisms for the production of vortices in helium droplets with about $N = 10^8$ atoms were first discussed in Ref. 250, as well as some methods for their detection based on the electronic spectroscopy of alkali atoms trapped on the vortex core. This idea has been further developed in Ref. 24, and will be reviewed below.

Detecting quantized vortices in He droplets still remains an open question. No unambiguous signature of their existence has yet been reported, and the whole high resolution rovibrational spectra of embedded molecules can apparently be explained without invoking their presence. This is somewhat deceiving; on the one hand, one of the common mechanisms for vortex creation in liquid helium, namely energy and angular momentum deposition by swift impurities, takes place naturally in experiments with doped droplets in molecular beams. On the other hand, it has been shown that impurities pin the vortex lines in droplets[251,252], so they might guarantee stability for a time long enough to permit detection.

The equilibrium configurations of a vortex line in a rotating superfluid drop were determined in Ref. 253. These authors also determined the potential barrier to vortex nucleation. More recently[254], these calculations have been extended to cold nanodroplets, in the context of vortex nucleation and stability in droplet jets.

A vortical state is an excited state of the superfluid, since the energy of droplets are significantly larger when hosting a vortex[251,253]. This, together with the failure to detect vortices, has been considered as a signature of their unstability in helium nanodroplets, since a newly created vortex could be rapidly expelled. However, it has been shown that vortices have considerably less excitation energy per unit angular momentum than the final states accessible by decay through the excitation of surface modes, the relevant ones for this process[254]. This is an interesting remark, as it indicates that droplet vortical states are the lowest possible states with high angular momentum, and this should hinder their decay.

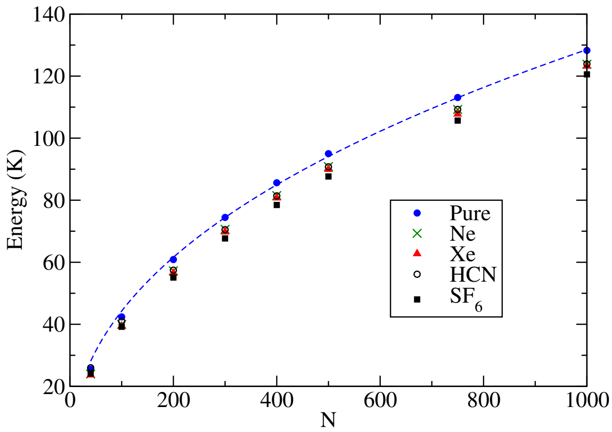

Fig. 29. (Color online) The vortex energy $\Delta E_{\rm V}(N)$ (dots) in pure $^4{\rm He}_N$ drops. The dashed line is a fit obtained using Eq. (20). Other symbols represent the vortex energies $\Delta E_{\rm V}^{\rm X}(N)$ in some doped droplets.

A detailed calculation of the vortex energetics in ^{4}He and mixed helium clusters, either pure or doped, has been presented in Refs. 251,252 (see also Ref. 255) using a DF approach and the Feynman-Onsager *ansatz*. In Fig. 29 we plot the energy associated with the vortex flow, defined as

$$\Delta E_{\rm V}(N) \equiv E_{\rm V}(N) - E(N) \; , \tag{19}$$

where $E_{\rm V}$ and E are the energies of droplets with and without vortex, respectively. The dashed line is the result obtained with a liquid drop like formula:

$$\Delta E_{\rm V}(N) = \alpha N^{1/3} + \beta N^{1/3} \log N + \gamma N^{-1/3} \ , \tag{20}$$

with the parameters α= 2.868 K, β= 1.445 K, and γ= 0.313 K extracted from a fit to the DF calculations. This formula, which works very well, is inspired in a hollow-core model for the vortex[251].

The next step is the inclusion of a dopant. The energetics of the system can be conveniently analyzed by introducing the energies

$$\Delta E_{\rm V}^{\rm X}(N) \equiv E_{\rm X+V}(N) - E_{\rm X}(N) \tag{21}$$

$$S_{\rm X+V}(N) \equiv E_{\rm X+V}(N) - E(N) \ , \tag{22}$$

where the subscripts X and V respectively refer to impurity and to vortex line. The energy excess $\Delta E_{\rm V}^{\rm X}$ is associated with the vortex flow in the doped cluster, some of which are shown in Fig. 29.

Two key quantities are the solvation energy of the dopant+vortex complex given by $S_{\rm X+V}(N)$ in Eq. (22), and the difference

$$\delta_{\rm X}(N) = \Delta E_{\rm V}^{\rm X} - \Delta E_{\rm V} \ . \tag{23}$$

When $\delta_{\rm X}$ is negative, its absolute value represents the binding energy of the dopant to the vortex in the cluster.

Since the solvation energy $S_{\rm X}$ is negative while the vortex energy $\Delta E_{\rm V}$ always increases, the dopant+vortex complex has a solvation energy S_{X+V} which changes sign at some $N_{\rm cr}$ which is within the range of droplet sizes in the experiments. This means that for $N < N_{\rm cr}$, the dopant+vortex complex is energetically favored. It turns out that for large enough droplets, $\delta_{\rm X}$ is -5.0 K, -4.4 K, and -7.7 K, for Xe, HCN, and SF_6, respectively[251]. Consequently, a fairly stable dopant+vortex complex could be formed by picking up the impurity, assuming that the collision imparts sufficient angular momentum. The vortex line is expected to appear attached to the dopant, since $\delta_{\rm X}$ is negative.

The structure of mixed $^3{\rm He}_{N_3}$+$^4{\rm He}_{N_4}$ droplets with an embedded impurity pinning a quantized vortex line has been also investigated[252]. It has been found that the dopant+vortex+$^4{\rm He}_{N_4}$ complex is robust against the addition of ^{3}He atoms. While the latter are distributed along the vortex line and on the surface of the ^{4}He drop, the impurity is mostly coated by ^{4}He particles, as seen in Fig. 30.

The failure to observe vortex lines in helium nanodroplet isolation spectroscopy has been addressed in Ref. 42. In this reference, it is argued that under present experimental conditions, the capture of impurities might deposit

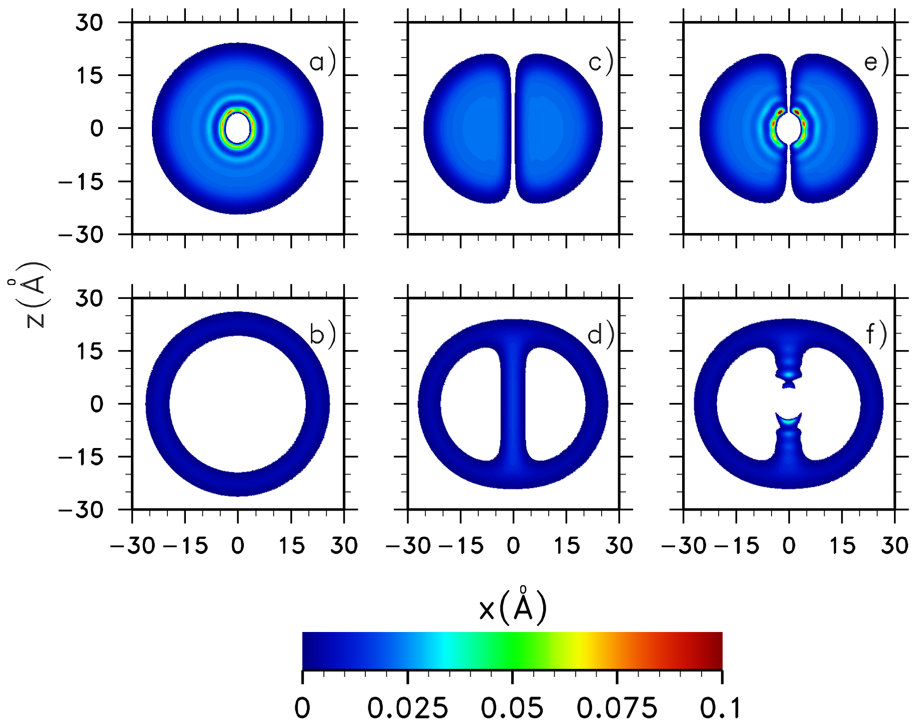

Fig. 30. (Color online) Grey scale plots of the ^{4}He (top panels) and ^{3}He atomic densities (bottom panels) for a $^4He_{500}$+$^3He_{100}$ mixed drop. The panels (a) and (b) show the densities of the drop hosting a linear HCN molecule aligned along the z-axis. It can be seen that ^{4}He atoms lie distributed into solvation shells around the molecule, whereas ^{3}He atoms are at the surface occupying Andreev states without disturbing the molecular superfluid environment. The panels (c) and (d) show the densities in the presence of a vortex along the z-axis. It can be seen that ^{4}He atoms are expelled from the vortex core, which is filled with ^{3}He atoms. The panels (e) and (f) correspond to the case when the drop has both an HCN molecule and a vortex line coaxial to the z-axis.

enough angular momentum into the droplet to form a vortex, but adding at the same time far more energy than required. For this reason, only a minute fraction of droplets might host a vortex, whereas for most droplets the energy and angular momentum is in the form of ripplon excitations[254,256]. Yet, we expect that in the near future, vorticity in helium nanoclusters is created by some extension of the present experimental techniques. This calls for identifying signatures of vortical states in helium droplets.

A dopant molecule in the bulk of the droplet has been considered a potential probe to detect vorticity, as both the density distortion near the pinning points, and the symmetry breaking arising from the existence of a vortex line, change the effective moments of inertia of the molecule, which are measurable by the changes they induce in the dopant rovibrational spectrum[2,4,30]. These changes are expected to be especially pronounced for linear molecules pinned along the vortex axis, whose rotation around an axis perpendicular to it will be either partially or completely suppressed, depending on the strength of the pinning forces. Unfortunately, the experiments do no show any evidence of the presence of vortices.

An experiment to detect vortices in nanoclusters has been proposed by Close et al.[250]. Their idea is that alkali atoms, that normally reside in a 'dimple' on the surface of ^{4}He clusters[161,204,208], may be drawn, when a vortex is present, *inside* the cluster along the vortex core. Electronic spectroscopy experiments[5] could thus provide evidence of their existence, since the line broadenings and shifts would be different in the two cases. However, it has been shown that alkali atoms are actually not suited for such an experiment[24]. Indeed, alkali atoms have too a robust stable state on the surface of liquid ^{4}He[161,208], and for these impurities the surface state is always preferred, even in the presence of a vortex line. Using the DF approach, a fully quantized vortex state has been generated starting from a pure or doped ^{4}He cluster[24], taking Na and Rb as representative of light and heavy alkalis, respectively. The stable 'dimple' states of Na and Rb atoms on the surface of the $N = 300$ cluster hosting a vortex line have been compared with those of the same impurity trapped in the vortex core, exactly at the cluster center. It has been found that the latter are energetically unfavored with respect to surface states.

To use atomic impurities as probes of the presence of vortices in ^{4}He drops, ideally one would like to have an atom that is *barely* stable on the surface of a pure drop, and becomes solvated in its interior in the presence of a vortex. There are other dopants for which there is clear evidence of a surface state on liquid ^{4}He, i.e. alkaline earth atoms. Measured absorption spectra of Ca, Ba and Sr attached to ^{4}He clusters clearly support an outside location of Ca and Sr[206], and probably also of Ba[257]. It has been found[24] that rather than alkalis, alkaline earth Ca atoms may serve as probes to detect vortices along the ideas put forward in Ref. 250.

The 'dimple' stable state of a Ca atom in a ^{4}He$_{300}$ vortex-free cluster is shown in Fig. 31(b). However, in the presence of a vortex, the stable state is in the center of the cluster, as depicted in Fig. 31(c). The surface state in this case is unstable and, as the minimization proceeds, a Ca atom initially placed on the surface near the vortex core, is gradually drawn towards it and then sucked inside, eventually reaching the stable state in the bulk of

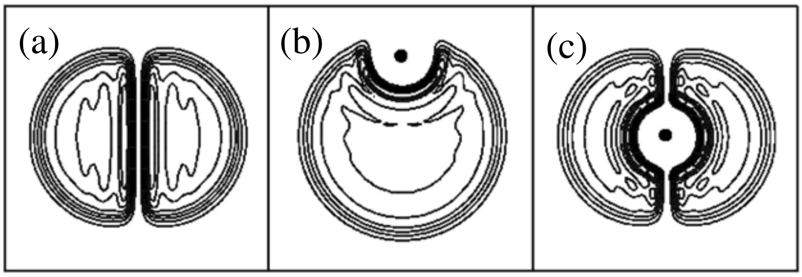

Fig. 31. From left to right, for a $^4He_{300}$ cluster, atomic equidensity lines corresponding to: (a) the vortical state; (b) the deep 'dimple' stable state of Ca, and (c) the solvated stable state of Ca in the presence of a quantized vortex. Each box side has 45 Å length.

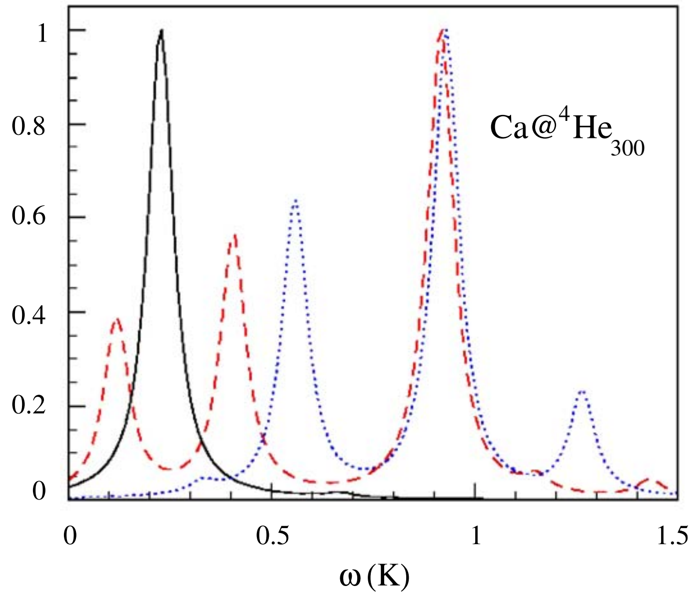

Fig. 32. (Color online) Frequency spectra for the oscillations of a Ca atom in a $^4He_{300}$ cluster. The intensities are in arbitrary units, and normalized so that the higher peak in each spectrum has unit height. Solid line: solvated state [see Fig. 31(c)], Ca moving along the vortex core; dashed line: solvated state, Ca moving perpendicular to the vortex core; dotted line: surface state [see Fig. 31(b)], Ca moving in the radial direction.

the drop.

The calculated vibrational spectra of Ca moving with respect to the cluster is reported in Fig 32 –to be not confused with the electronic spectrum of Ca. It appears that the oscillation of Ca along the vortex core is characterized by a single low frequency mode, as compared with the more fragmented spectrum for perpendicular vibrations. The presence of the soft mode is a signature of solvation of a Ca atom inside the vortex. Indeed, such a mode should be severely damped, or even absent, for a solvated Ca atom in a vortex-free cluster, since this configuration is unstable. We also show for comparison the vibration spectrum of a Ca atom in the 'dimple' state on the surface of a cluster without vortex. The peak just below 1 K is due to the 'dipolar' vibration of the impurity inside the semi-spherical(spherical) cavity in which it is trapped in the 'dimple' state. Additional peaks appear in the 'dimple' spectra because of the coupling of the Ca motion to the surface modes of the ^{4}He nanodroplet, which have similar frequencies. Note that all these modes lie in the microwave frequency regime, and their detection calls for new experimental ideas.

Our calculations confirm that the dynamical behavior of cluster-dopant systems depends in a very sensitive way on the details of the He-atom pair interaction. This calls for improvements on the available interaction potentials; research in this direction might be helpful as well to find other atomic impurities which may serve as probes of the presence of vortices in ^{4}He droplets.

7. HELIUM DROPLETS ON ADSORBING SUBSTRATES

When a helium cluster is placed into an external field $V(\mathbf{r})$ each atom experiences, in addition to the interaction with the environment, a force $\nabla V(\mathbf{r})$. The restoring force in the vicinity of the potential minimum confines the atoms to that neighborhood, changing in an appreciable manner the spatial distribution and the energetics of the system. One of the desirable effects of confinement is the possibility of stabilizing thermodynamically unstable configurations; for example, slabs of helium with finite spread along the z-dimension, which are unstable against addition of material[258], may become stable films when adsorbed into the potential $U_s(z)$ created by a sufficiently strong planar substrate, placed near one of the slab surfaces.

The wetting properties of liquid helium have been widely investigated. Experiments have collected data on adsorption isotherms, interfacial surface tensions, wetting temperatures and phase diagrams for the pure liquids and mixtures, as reviewed i.e., in Refs. 259–263. More than one decade ago, it was shown, both experimentally and theoretically, that liquid ^{4}He is not a universal wetting agent, since it does not condense into a thick uniform film on cesium surfaces at temperatures below 1.95 K. By contrast, ^{3}He has been shown to be a universal wetting agent[264,265]. This is due to Fermi statistics, that forces the chemical potential of ^{3}He to increase as the size of the system grows, so that the Fermi energy of the adsorbed particles approaches the binding energy of a single ^{3}He atom on the planar surface[263,264]. On the other hand, it is well established that ^{3}He populates a two-dimensional homogeneous layer of Andreev states on the free surface of bulk ^{4}He and films (see e.g., Ref. 266 and references therein); in a low-temperature dilute solution, the Fermi pressure exerted by the ^{3}He impurities at the surface reduces the liquid-vapor and liquid-solid surface tensions of ^{4}He. This permits lower wetting temperatures as the ^{3}He concentration increases[267].

The majority of theoretical works on wetting properties of helium focus upon structure and energetics of planar uniform films, parallel to the substrate plane. In particular, DF theory was able to predict the appearance of Andreev-like states localized at the liquid-substrate interface of a ^{4}He film on a variety of adsorbers, including weak alkalis like Cs[268–270]. The thermodynamic criterium for zero-temperature wetting is expressed in terms of the surface grandpotential $\sigma = (E - \mu N)/A$, where A is the area of the surface, as a function of areal coverage $n = N/A$. A surface is wetted if the absolute minimum occurs when n approaches infinity[271,272]. From the thermodynamic relations defining the chemical potential $\mu = e + n\dfrac{\partial e}{\partial n}$ with $e = E/N$, and the surface tension

$$\sigma = -n^2\,\frac{\partial e}{\partial n} = \int_0^n dn'\,[\mu(n') - \mu(n)] \tag{24}$$

the condition $\mu = e$ –or equivalently, the vanishing of the grandpotential– localizes the prewetting-like first order transition at zero temperature through a Maxwell construction. This is illustrated in Fig. 33, where we can see the chemical potential of ^{4}He, computed with the OT density functional plus the Chizmeshya-Cole-Zaremba (CCZ) substrate potential[273], on different alkali surfaces as a function of coverage, as well as the horizontal Maxwell plateau.

In this figure, ^{4}He wets K but not Cs and Rb surfaces[274]. For the lowest areal coverages, these systems are thermodynamically unstable and form puddles or clusters on the surface, even in the case of wetted adsorbers[259]. DF methods have been applied to investigate mixed clusters on adsorbing substrates, and one of the most important results is the evidence that the lowest-lying ^{3}He sp bound states are localized on the contact line that surrounds the ^{4}He base[260,263]. These are genuine edge states of a different nature than the well-known Andreev ones, since they originate in the peculiarities of the mean field for this specific geometry. A detailed study of the traditional Andreev states of planar films as functions of the ^{4}He coverage permits to establish that edge states originate in the finite geometry

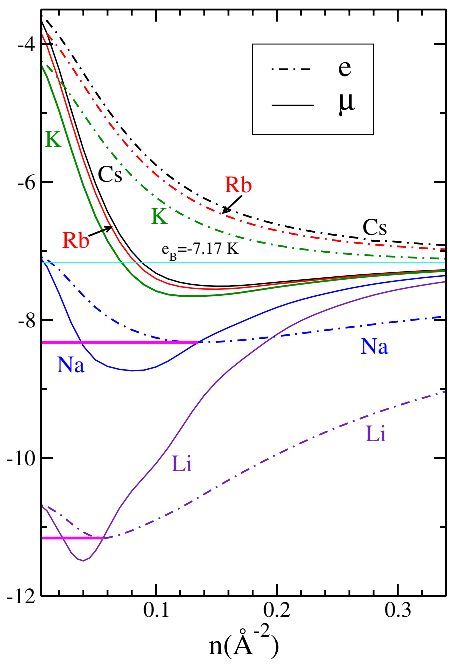

Fig. 33. (Color online) Chemical potential μ (K) of ^{4}He films on Cs, Rb, K, Na and Li substrates at zero temperature, as functions of areal coverage n (Å^{-2}). Also shown is the energy per particle in bulk ^{4}He e_B as a thin straight line. The thick horizontal segments correspond to the Maxwell construction that locates the prewetting jump in Na and Li.

of the mixed cluster, since they appear even when Andreev states are unbound on the prewetting film. One can also investigate the sp spectrum of ^{3}He atoms in $^3\mathrm{He}_{N_3}$+$^4\mathrm{He}_{N_4}$ mixed clusters, predicted by the solution of the KS equations derived from the DF, as well as on the shapes of both helium distributions, as functions of both sizes.

7.1. $^4\mathrm{He}_N$ nanodroplets on planar alkali substrates

Systematic studies of droplets deposited on surfaces carried in the DF frame[261] permit to relate the asymptotic trends of cluster energetics and structure for increasing size, to the wetting properties of the fluid. The description is achieved by solving an Hartree-like equation for $\sqrt{\rho(\mathbf{r})}$ whose field includes the substrate potential generated by the alkali substrate. The first calculation of density profiles of finite droplets of ^{4}He on Cs, at zero temperature, was performed by Ancilotto *et al* in Ref. 69, employing the OT functional and the CCZ He-alkali potential. In this study, density profiles $\rho(\mathbf{r})$ of axially symmetric clusters flattened on a planar Cs surface were presented together with their energetics. This procedure was later extended[261] to investigate the growth of ^4He_N clusters on alkali adsorbers.

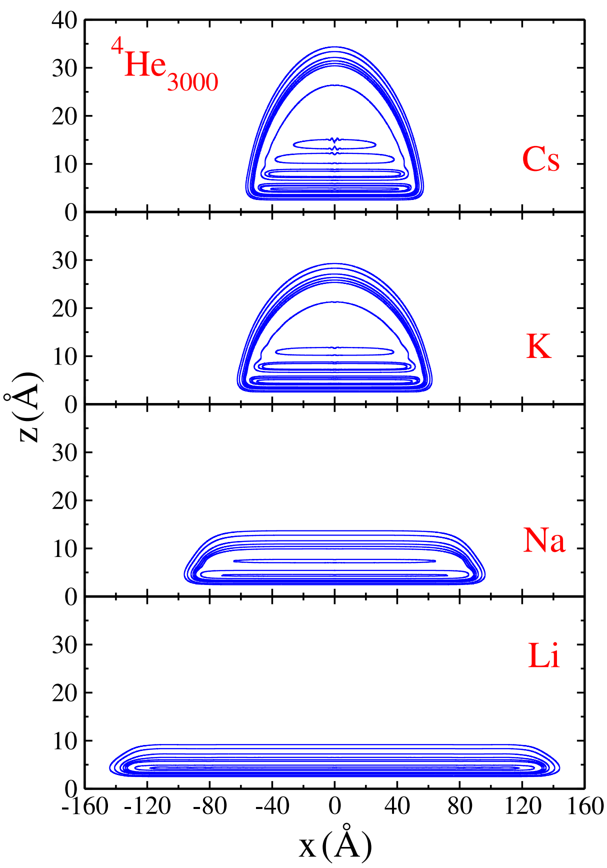

Fig. 34. (Color online) Density profiles $\rho_4(x, z)$ of $^4He_{3000}$ droplets on Cs, K, Na and Li.

The typical structure of these clusters is illustrated in Fig. 34, where we see the contour plots of the particle densities in the (x, z) plane for a cluster with $N = 3000$, on substrates Cs, K, Na and Li. The change of shape is clear;

the stronger the attraction, the more flattening along the vertical coordinate accompanied by radial spreading, most noticeable for the clusters on Na and Li. The structure displays three ridges on Cs and K, two on Na, and one on Li. This reflects the prewetting behavior of ^{4}He that occurs with jumps of above three layers on K, two on Na and one on Li, respectively. These features are confirmed by the analysis of the density profiles sliced along the direction perpendicular to the alkali substrate[261]. It can be seen that even for Li, these drops have some transverse structure. Strictly two-dimensional puddles of liquid ^{4}He have been addressed within shadow variational wave function[275], DMC[276], and DF approaches[277].

The chemical potential of the ^{4}He particles, and the grandpotential per particle $\omega = E/N - \mu$ as functions of the number of atoms, can be seen in Fig. 35. For wetted adsorbers, μ asymptotically approaches a finite value below the bulk value, while ω tends to vanish. This is very clear for the strongest Na and Li and occurs at N above 10^4 atoms for K. For Cs and Rb surfaces, the limiting chemical potential approaches the bulk figure from above, in agreement with the nonwetting behavior of ^{4}He on these substrates.

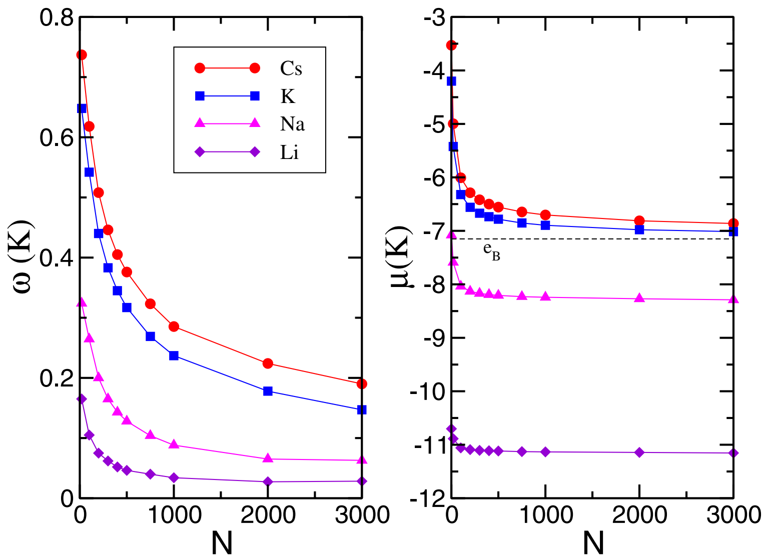

Fig. 35. (Color online) Chemical potential (right panel) and grandpotential per particle (left panel) of ^{4}He droplets on alkali adsorbers as functions of N. e_B in the right panel represents the energy per particle in bulk ^{4}He.

The cluster growth on a selected substrate is illustrated in Fig. 36, that shows contour plots of the particle density of ^{4}He droplets on Na for N between 100 and 3000. We observe that the matter distribution saturates with two shells and a limiting height, and flattens into a wedge-like shape. If one introduces an effective coverage

$$n(r) = 2\,\pi \int dz\, \rho(r,z) \tag{25}$$

the examination of $n(0)$ as a function of N for all substrates under consideration shows that for wetted adsorbers, asymptotic values of the central effective coverage –and consequently, of the chemical potential– exist. In Ref. 261 the values $(\mu_{Na}, n_{Na}(0))$ = (-8.27 K, 0.14 Å^{-2}) and $(\mu_{Li}, n_{Li}(0))$ = (-11.15 K, 0.06 Å^{-2}) have been encountered, which coincide with the respective prewetting values in Fig. 33.

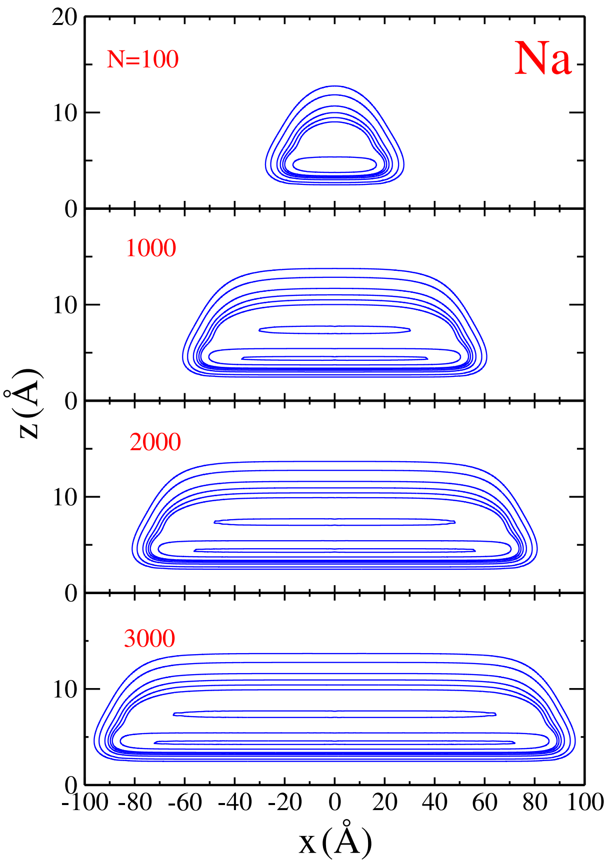

Fig. 36. (Color online) Contour plots of the density of a ^4He_N droplet on Na for N = 100, 1000, 2000, and 3000.

7.2. Edge states in ^{3}He + ^{4}He$_N$ clusters

We now review DF calculations of nanoscale mixed ^{3}He-^{4}He$_N$ droplets at zero temperature on a flat Cs surface[260,262,263], which are the only existing calculations. The philosophy is to solve the Schrödinger equation for the ^{3}He atom in the external field of an alkali surface plus a ^{4}He$_N$ drop on the same substrate, for a sp wave function $\phi_\alpha \equiv \phi_{nl}$

$$\phi_{nl}(r,z,\varphi) = \psi_{nl}(r,z)\,\frac{e^{il\varphi}}{\sqrt{2\pi}}\,, \tag{26}$$

l being the z-projection of the sp orbital angular momentum, and energy ε_{nl}. For large N values, the states (nl) with $n=0$ and $l=0,1,2,\ldots$ are distributed into rotational bands

$$\varepsilon_{0l} = \varepsilon_{00} + \frac{\hbar^2 l^2}{2m^*_{00}R^2_{00}} \tag{27}$$

where the effective masses m^*_{nl} are defined in Refs. 260,263.

The probability distribution functions $|\psi_{0l}(r,z)|^2$ are essentially identical to $|\psi_{00}(r,z)|^2$, revealing that these excitations consist of rotations of the gs ψ_{00} around the symmetry axis of the ^{4}He drop. On the other hand, the probability densities $|\psi_{n0}|^2$ of s-excited states ψ_{n0} do not vanish on the z-axis; thus, when $l>0$, the centrifugal potential splits the cusp and creates pairs of lobes. If N is small, say a few tens of atoms, the energy levels do not group into rotational bands for any n, due to the fact that the probability densities $|\psi_{nl}|^2$ lie too close to the z axis and cannot avoid centrifugal distortion at any nonvanishing l.

In Fig. 37 we can see contour plots of the gs probability density $|\psi_{00}(r,z)|^2$ on the $y=0$ plane, together with the density $\rho_4(r,z)$ of the cluster ^{4}He$_{3000}$ on Cs. The gs represents a new kind of sp state, strongly localized on the contact line, that has been named the edge state[260]. The probability density is 1D on a ring of mesoscopic radius. Its origin is the combination of the density distribution of the ^{4}He drop, which in turn creates the mean field experienced by the ^{3}He particle, and the potential generated by the substrate, with a strong attraction for the impurity near the contact line.

A systematic investigation of the different alkali substrates for varying host size showed that the existence of the edge state is a universal feature. The gs is always localized on the contact line and for $N_3>1$, the probability densities display several fringes on the surface of the ^{4}He cluster, as found in Ref. 260 for the case of a Cs adsorber. If N_3 is of the order of a few tens of atoms, the edge state is still present, but excited ^{3}He probability densities are pulled up into unbound configurations. The number of ψ_{n0}

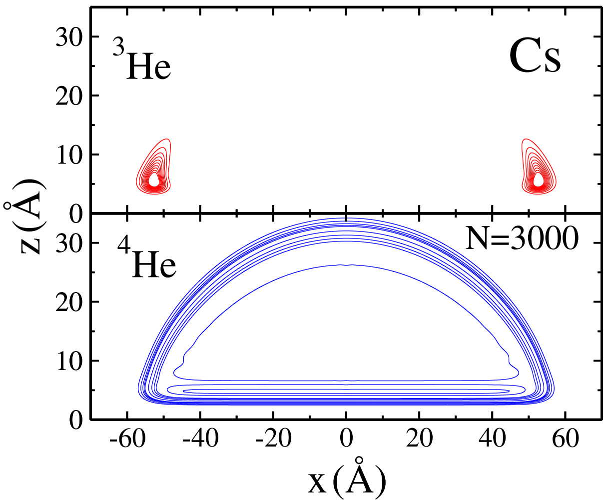

Fig. 37. (Color online) Probability density of the gs of a ^{3}He atom (upper panel) together with the ^{4}He$_{3000}$ particle density in the $y = 0$ plane on a Cs surface.

localized bound states, with energies below that of a single ^{3}He atom on a planar surface, $\mathcal{E}_3$, varies both with N_4 and the alkali substrate; for example, droplets larger than ^{4}He$_{100}$ on K and on Cs support three bound $s-$states. Instead, for Na and Li only one single $l = 0$ bound state has been found, the edge state.

The energetics of the ^{3}He atom across the prewetting transition deserves a few comments. For the K adsorber, the weakest of this series, the impurity spectrum is essentially constant with coverage; for K, Na and Li the first excited state appears above the gs of the ^{3}He atom on the substrate. In contrast, at the prewetting density n_c the energy of the Andreev state lies below $\mathcal{E}_3$ for K, and above for Na and Li. This is an important finite size effect, since on these adsorbers, the edge state is favored with respect to binding to the substrate and permits the deposition of mixed drops. By contrast, on adsorbers like Na and stronger, when the edge-like states are filled, the probability density of any extra ^{3}He atom spreads on the alkali surface.

7.3. ^{3}He$_{N_3}$+^{4}He$_{N_4}$ clusters

Mixed helium droplets on alkali surfaces at $T = 0$ have been discussed in Ref. 263. The outcome of these calculations is illustrated in Fig. 38, where we can see the lowest KS sp energies ε_{nl} as functions of N_3 for ^{4}He$_{20}$

on Cs. Heavy squares indicate the ^{3}He Fermi energy (or chemical potential μ_3) of each configuration, clearly a nonmonotonically increasing function of N_3, as encountered in homogeneous planar films[264,268,270]. Strong shell closures yield magic numbers for the fermion component, depending both on the alkali substrate and on N_4. Contour plots of the particle densities $\rho_3(x,z)$ and $\rho_4(x,z)$ are shown in Fig. 39 for $N_4 = 20$, and two N_3 values. These patterns illustrate that the density profiles ρ_3 spread on the surface and tend to cover the horizontal base, as N_3 grows. Slight dilution inside the bulk of the host can be also appreciated. One also realizes that ^{3}He atoms in Andreev surface states coat the ^{4}He cap, which remains fairly insensitive to the presence of ^{3}He and does not spread on the substrate surface. This behavior can be understood in the light of the sp spectrum in Fig. 38; since the Fermi energy of the ^{3}He atoms increases with N_3, it is apparent from this figure that for N_3 above 44, it is energetically more convenient for ^{3}He particles to spread on the substrate, than to remain in the vicinity of the ^{4}He drop.

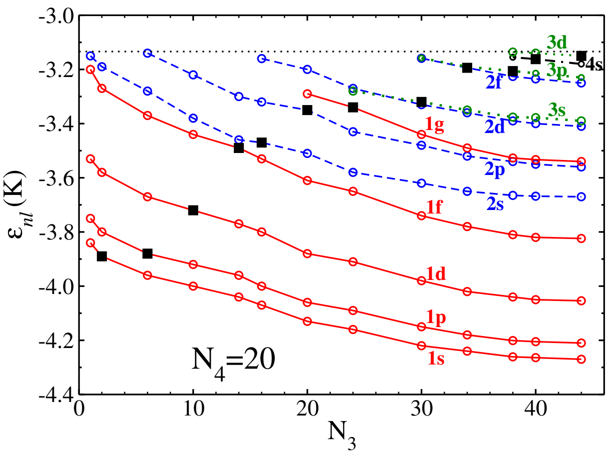

Fig. 38. (Color online) The Kohn-Sham sp energies ε_{nl} for $^3\mathrm{He}_{N_3}$+$^4\mathrm{He}_{20}$ mixed drops on Cs as functions of N_3. The horizontal line indicates the energy $\mathcal{E}_3$ of one ^{3}He atom in the Cs adsorbing field. Heavy squares indicate the corresponding ^{3}He Fermi energy. The lines connecting the sp energies of (nl) states have been drawn to guide the eye.

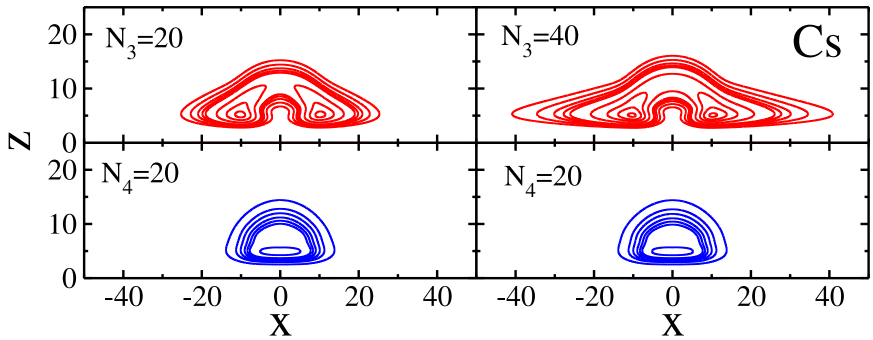

Fig. 39. (Color online) Contour plots of the density profiles $\rho_3(x,z)$ (top panels) and $\rho_4(x,z)$ (bottom panels) of $^4\mathrm{He}_{20}$+$^3\mathrm{He}_{N_3}$ mixed drops on Cs, for N_3 = 20 (left panels) and 40 (right panels). In the case of ^{4}He, the equidensity lines correspond to $10^{-4}, 10^{-3}, 2.5\times10^{-3}, 5\times10^{-3}, 7.5\times10^{-3}, 10^{-2}, 2\times10^{-2}$, and 2.5×10^{-2}Å^{-3}, and in the case of ^{3}He, they correspond to $10^{-4}, 2.5\times10^{-4}, 5\times10^{-4}, 7.5\times10^{-4}, 10^{-3}, 2.5\times10^{-3}, 5\times10^{-3}, 7.5\times10^{-3}$, and 10^{-2}Å^{-3}.

7.4. Growth instability in the presence of extended sources

It has been recently shown that the physics of doped droplets may be linked to the physics of wetting, due to the possibility of capturing atomic clusters with sizes comparable to the drop size. This view has been applied to study the growth of helium shells on carbon fullerenes, nanoscopic metallic spheres or spherical sheets a few Å radii[278–281]. In these cases, the impurity presents a curved surface on which the helium atoms may condense, undergoing a prewetting-like transition, and form spherical shells. In agreement with earlier findings regarding wetting of curved surfaces by classical liquids, such systems present a growth instability that prevents stable approach to the bulk condition beyond a maximum shell thickness. This feature may be compared with the growth pattern of droplets in the presence of point sources as discussed in Sec. 4. Earlier investigations of wetting of curved surfaces by classical liquids (see e.g., Ref. 282 and cited therein) have demonstrated that the areal energy balance involving a free liquid-vapor or liquid-vacuum interface, larger than the liquid-solid one, sets an upper limit to the thickness of an adsorbed spherical or cylindrical film. This can be understood on thermodynamical grounds, since the wetting condition is given by the energy balance[283]

$$A_R\,\sigma_{sl} + A_{R+d}\,\sigma_{lv} = A_R\,\sigma_{sv}\,, \tag{28}$$

where A_R, A_{R+d} respectively denote the area of a spherical or cylindrical solid substrate of radius R and that of a sphere or cylinder of radius $R+d$ at the liquid-vapor interface with a film of thickness d. Here σ_{ij} is the surface tension between phases i and j, where labels denote solid (s), liquid (l) or vapor/vacuum (v). In this situation, geometry and thermodynamics are entangled, since from the energy balance one immediately gets

$$\sigma_{sl} + \sigma_{lv}\left(1 + \frac{d}{R}\right)^2 = \sigma_{sv}\,, \tag{29}$$

so that for any finite radius R and temperature, the film size d cannot grow up to infinity. Accordingly, continuous wetting is unattainable on a curved substrate.

A detailed investigation of the adsorption pattern of $^4\mathrm{He}_N$ on carbon fullerenes[278] and metallic spheres[279] employing a DF to extract density profiles and energetics, confirmed the existence of a growth instability that prevents stable approach to the bulk condition beyond a maximum shell thickness. This can be seen in Fig. 40 where the ^{4}He chemical potential μ is plotted as a function of N for fullerenes C_{20} and C_{60}. An overshoot of the

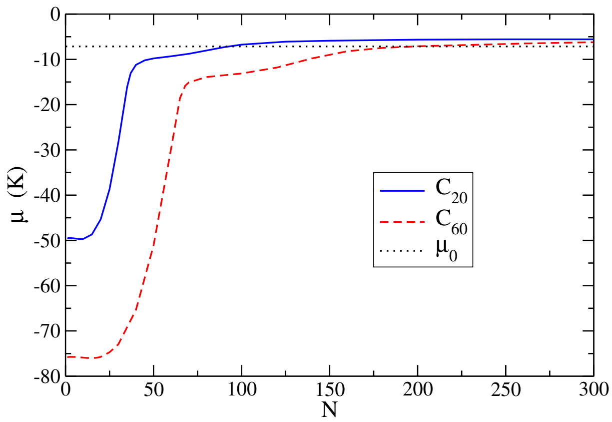

Fig. 40. (Color online) Chemical potential of ^{4}He on fullerenes C_{60} and C_{20} as a function of atom number N. The ^{4}He bulk chemical potential μ_0 is also shown.

trends $\mu(N)$ is visible: at particle numbers $N_0 = 93$ for C_{20} and 198 for C_{60}, the chemical potential of the helium shell crosses the bulk value μ_0, and reaches a broad maximum before beginning a smooth approximation to μ_0 with negative slope. Spherical shells with more than N_0 atoms are unstable against further addition of material.

The growth instability is intrinsic to the existence of a finite curvature $1/R$ [cf. Eq. (29)] and it is interesting to note that it is already present in the growth patterns of $^3\mathrm{He}_N$ and $^4\mathrm{He}_N + X$ discussed in Sec. 4. The curves

$\mu_4(N)$ in Fig. 2 of Ref. 230 and $\mu_3(N)$ in Fig. 4 of Ref. 135 show that the chemical potential of the nanoclusters undergo an overshoot and approach the bulk value with a negative slope.

The instability discussed above can be removed by capillary condensation in a sufficiently wide spherical gap[281]. In such a configuration, the limit to the amount of storable material is set by the capacity of the cavity, unless an extra pressurizing field is applied to the fluid.

8. SUMMARY

In this paper we have revised various aspects of the current knowledge on helium nanoclusters, focusing on theoretical interpretations of available data and on predictions with possible experimental confirmation. Rather than being exhaustive in the collection of information, we have pursued a twofold purpose. On the one hand, we have pointed out several features of the physics of helium nanodroplets that stress their unique character and illustrate their richness, both from the theoretician's and from the experimentalist's viewpoints. On the other hand, we have emphasized that descriptions of these helium systems based on the use of the bare He-He interaction, and DF descriptions based on the use of an effective interaction supplemented with terms that phenomenologically include correlations are complementary, since each of them aims at covering a specific domain of systems and of properties. In the overlap region consisting of moderate cluster sizes, optionally including dopants, the exact calculations performed, i.e. with QMC or PIMC methods, make room to confidence in DF results and predictions beyond this region.

Microscopic calculations on helium complexes ranging from dimers up to clusters with a few tens of atoms call our attention on the importance of the helium-helium interaction, by far a nontrivial issue; it should be recalled that apparently minor differences in the appearance and parameters of the force give rise to a factor two in the binding energy of the ^{4}He dimer. We can also understand the role of the different isotopic masses and the statistics undergone by the atoms by analyzing the energetics of tiny clusters. We have seen that a microscopic treatment, as well as a DF-plus-shell-model approach, may shed light on the minimum number of ^{3}He atoms that can be self-bound, and on the spin structure of open shell ^{3}He droplets.

We have stressed the fundamental role of QMC methods in our current understanding of the coupling of a dopant to its helium environment, as manifested by the satisfactory description of the changes in the moment of inertia of the molecule. Important achievements of these microscopic descriptions are the neat localization of the superfluid fraction in such rotator-helium complexes, the comprehension of the changes in the rotational constant of the molecule with drop size, and the description of the transition between the weakly bound helium-impurity configuration, and a scenario with a quantum solvated molecule in the bulk of the host.

We have shown as well that DF theory may provide robust interpretations of various measured properties that involve large numbers of He atoms of either isotope in the cluster, and intriguing evidence of some phenomena which still remain unobserved, e.g. the structure of pure and doped drops hosting a vortex, and the demonstration that the gs of ^{3}He atoms in ^{4}He drops on alkali substrates is a true edge state, located along the drop-substrate contact line. Finally, we have revised our current knowledge of isotopically pure and of mixed helium drops on alkali substrates, that constitutes a novel perspective of the physics of wetting by liquid helium, traditionally based on the description of films.

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